

Guidance on Lecture Slides Package and Pre-Read Activity Recording

Dear Students

Please note that the slide provided in the **Pre-Read Activity Recording** contains, to a large extent, the same slide content as that of the slide deck found in the **Lecture Slides** package. However, the order of the slides deck in the **Lecture Slides** package may not be in sync with that of the slides found in the **Pre-Read Activity Recording**. Regardless, you can still use these slides to make notes for your lectures.

We will be using the slides found in the **Lecture Slides** package during lectures.

Not everything discussed in the **Pre-Read Activity Recording** will be covered in the lecture.



Basic Principles of Medicine 1

Module: Endocrine and Reproductive System | Lecture 01

Integrative Histology of the Endocrine Glands (Flipped Classroom)

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Assigned Reading

- **Pre-Read Slide Packet**

- Histology: A Text and Atlas: With Correlated Cell and Molecular Biology, 9e Wojciech Pawlina, Michael H. Ross
 - Chapter 21
 - Chapter 18 (Endocrine Pancreas)
 - Link to textbook <https://sgu.idm.oclc.org/login?url=https://premiumbasicssciences-lwwhealthlibrary-com.sgu.idm.oclc.org/book.aspx?bookid=3290>
- Additional Texts
- Wheater's Basic Pathology A text, Atlas and Review of Histopathology, 5e Barbara Young, William Stewart, Geraldine O'Dowd

Objectives

SOM.MK.I.BPM2.2.ER.1.HCB.1001	Classify endocrine glands based on chemical components of their secretion.
SOM.MK.I.BPM2.2.ER.1.HCB.1002	Describe the basic histological characteristics of all endocrine glands.
SOM.MK.I.BPM2.2.ER.1.HCB.1003	Describe the development of the pituitary gland in terms of its cellular and neuronal origins.
SOM.MK.I.BPM2.2.ER.1.HCB.1004	Identify histologically the glandular and nervous tissue components of the pituitary gland
SOM.MK.I.BPM2.2.ER.1.HCB.1005	List the components of the hypothalamohypophyseal tract.
SOM.MK.I.BPM2.2.ER.1.HCB.1006	Discuss the significance of the hypothalamohypophyseal tract in terms of hormone production and secretion.
SOM.MK.I.BPM2.2.ER.1.HCB.1007	List the vascular components of the hypothalamohypophyseal portal system.
SOM.MK.I.BPM2.2.ER.1.HCB.1008	Discuss the significance of the hypothalamohypophyseal portal system in terms of hormone production and secretion.
SOM.MK.I.BPM2.2.ER.1.HCB.1009	List the hormones secreted by the anterior pituitary
SOM.MK.I.BPM2.2.ER.1.HCB.1010	Describe the microscopic structures of anterior pituitary.
SOM.MK.I.BPM2.2.ER.1.HCB.1011	Briefly discuss the main effects of the hormones produced by the anterior pituitary on their target tissues.
SOM.MK.I.BPM2.2.ER.1.HCB.1012	Describe the location of pars intermedia.
SOM.MK.I.BPM2.2.ER.1.HCB.1013	Describe the microscopic structure of pars intermedia.
SOM.MK.I.BPM2.2.ER.1.HCB.1014	List the two (2) hormones that are stored in the posterior pituitary.

Objectives

SOM.MK.I.BPM2.2.ER.1.HCB.1015	Describe the microscopic structure of the posterior pituitary.
SOM.MK.I.BPM2.2.ER.1.HCB.1016	Give the origin of the two (2) hormones that are stored in the posterior pituitary.
SOM.MK.I.BPM2.2.ER.1.HCB.1017	Discuss the main effect(s) of the two (2) hormones that are stored in the posterior pituitary on their target tissue.
SOM.MK.I.BPM2.2.ER.1.HCB.1018	Discuss pituitary adenomas and possible clinical presentations.
SOM.MK.I.BPM2.2.ER.1.HCB.1019	Describe the anatomical location of the adrenal gland.
SOM.MK.I.BPM2.2.ER.1.HCB.1020	Describe the blood supply to the adrenal gland.
SOM.MK.I.BPM2.2.ER.1.HCB.1021	List the cortical zones of the adrenal gland.
SOM.MK.I.BPM2.2.ER.1.HCB.1022	Describe the microscopic structure (including the cell types) of the cortical zones of the adrenal gland.
SOM.MK.I.BPM2.2.ER.1.HCB.1023	Give the hormones produced in each of the cortical zones of the adrenal gland.
SOM.MK.I.BPM2.2.ER.1.HCB.1024	Briefly discuss the main effect(s) of the hormones in each of the cortical zones of the adrenal gland on their target tissue(s).
SOM.MK.I.BPM2.2.ER.1.HCB.1025	Identify the medulla of the adrenal gland.
SOM.MK.I.BPM2.2.ER.1.HCB.1026	Describe the microscopic structure of the medulla of the adrenal gland.
SOM.MK.I.BPM2.2.ER.1.HCB.1027	Give the hormone(s) produced in the medulla of the adrenal gland.
SOM.MK.I.BPM2.2.ER.1.HCB.1028	Briefly discuss the main effect of the hormones produced in the medulla of the adrenal gland on their target tissues.
SOM.MK.I.BPM2.2.ER.1.HCB.1029	Compare and contrast Cushing's disease vs. Cushing's syndrome.

Objectives

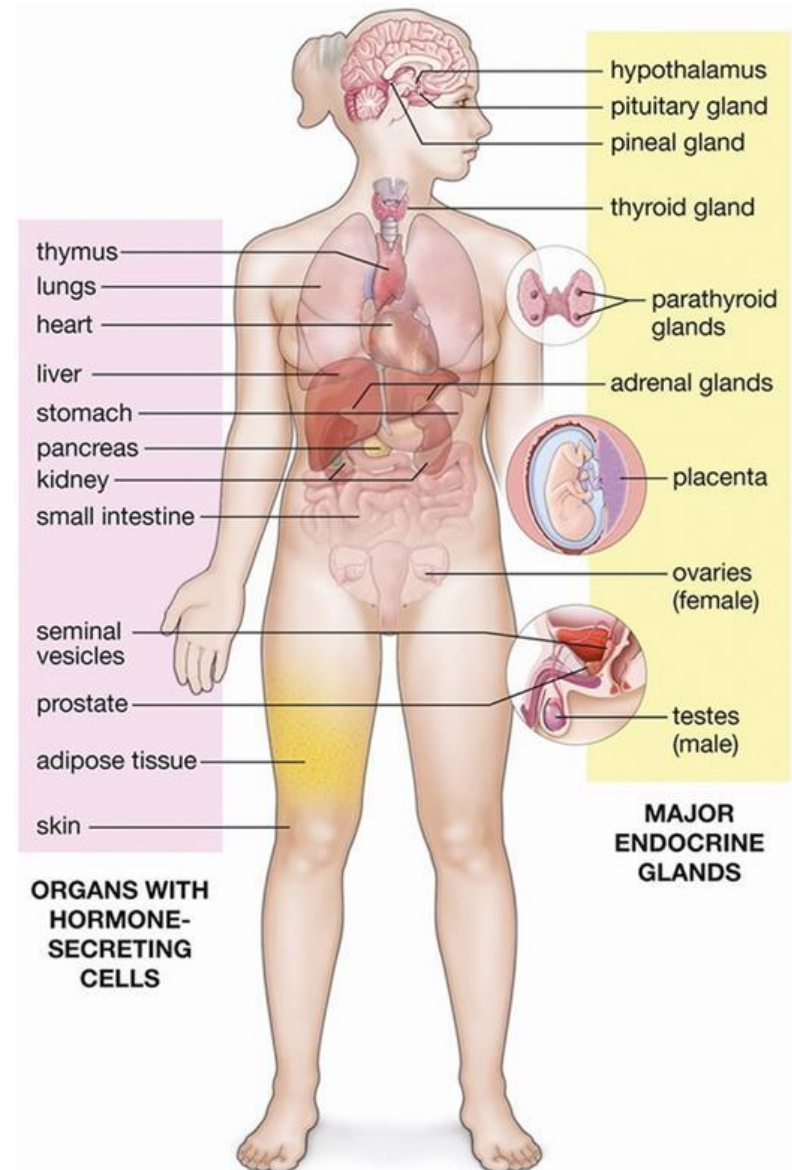
SOM.MK.I.BPM2.2.ER.1.HCB.1030	Describe the anatomical location of the pancreas.
SOM.MK.I.BPM2.2.ER.1.HCB.1031	Describe the blood supply to pancreas.
SOM.MK.I.BPM2.2.ER.1.HCB.1032	Briefly distinguish between the exocrine and the endocrine pancreas.
SOM.MK.I.BPM2.2.ER.1.HCB.1033	Identify the pancreatic islets of the endocrine pancreas.
SOM.MK.I.BPM2.2.ER.1.HCB.1034	List the cells of the pancreatic islets of the endocrine pancreas.
SOM.MK.I.BPM2.2.ER.1.HCB.1035	Discuss the microscopic structure of the pancreatic islets of the endocrine pancreas, including the location of each cell type found within the islets.
SOM.MK.I.BPM2.2.ER.1.HCB.1036	Briefly discuss the main effects of the hormones secreted by the cells of the pancreatic islets on their target tissues.
SOM.MK.I.BPM2.2.ER.1.HCB.1037	Describe the anatomical location of the thyroid gland.
SOM.MK.I.BPM2.2.ER.1.HCB.1038	State the function(s) of the thyroid gland.
SOM.MK.I.BPM2.2.ER.1.HCB.1039	Describe the microscopic appearance of the thyroid gland.
SOM.MK.I.BPM2.2.ER.1.HCB.1040	Describe the microscopic changes seen in the thyroid gland in hypothyroidism.
SOM.MK.I.BPM2.2.ER.1.HCB.1041	Describe the changes seen in the thyroid gland in hyperthyroidism.
SOM.MK.I.BPM2.2.ER.1.HCB.1042	Give the localization of parafollicular cells.
SOM.MK.I.BPM2.2.ER.1.HCB.1043	Discuss the function of parafollicular cells.
SOM.MK.I.BPM2.2.ER.1.HCB.1044	Describe the microscopic appearance of the parafollicular cells.

Objectives

SOM.MK.I.BPM2.2.ER.1.HCB.1045	Briefly discuss the main effect of the hormone secreted by the parafollicular cells on their target tissue(s).
SOM.MK.I.BPM2.2.ER.1.HCB.1046	Give the anatomical location of the parathyroid glands.
SOM.MK.I.BPM2.2.ER.1.HCB.1047	Discuss the function of the parathyroid glands.
SOM.MK.I.BPM2.2.ER.1.HCB.1048	Describe the microscopic appearance (including the 2 cell types) of the parathyroid glands.
SOM.MK.I.BPM2.2.ER.1.HCB.1049	Briefly discuss the main effect of the hormone secreted by the main cell type of the parathyroid glands on their target tissue(s).
SOM.MK.I.BPM2.2.ER.1.HCB.1050	Analyze the relationship between the parathyroid glands and the parafollicular cells.
SOM.MK.I.BPM2.2.ER.1.HCB.1051	Give the localization of the pineal gland.
SOM.MK.I.BPM2.2.ER.1.HCB.1052	Give the function(s) of the pineal gland.
SOM.MK.I.BPM2.2.ER.1.HCB.1053	Describe the microscopic structure (including the 2 cell populations) of the pineal gland.
SOM.MK.I.BPM2.2.ER.1.HCB.1054	Discuss the clinical significance of corpora arenacea.
SOM.MK.I.BPM2.2.ER.1.HCB.1055	Briefly discuss the main effect of the major hormone secreted by the pineal gland on its target tissue(s).

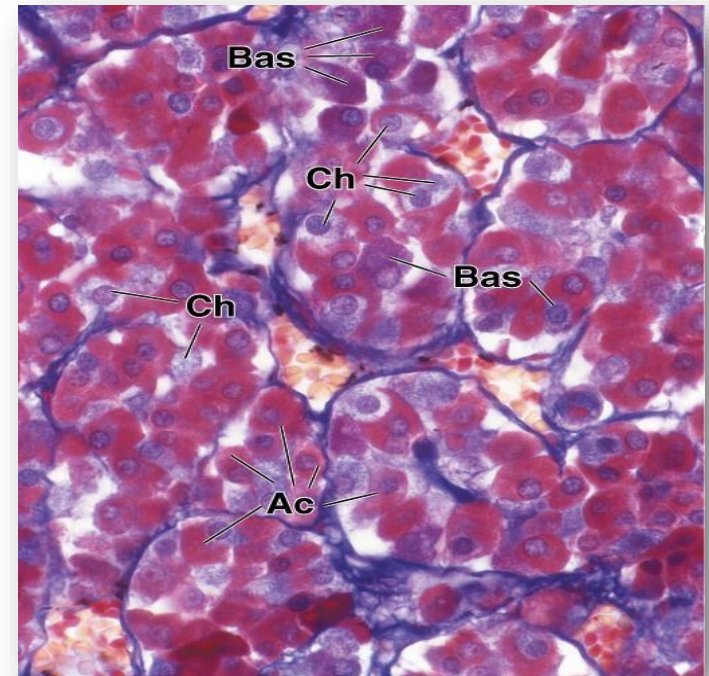
How does it work?

- Endocrine system has a crucial role in regulating several physiological processes and **maintaining homeostasis**.
- **Hormones are chemical messengers** that bind to specific receptors on target cells.
- Feedback systems: Positive or Negative.
- **Central Endocrine Glands:** associated with brain structures: **Hypothalamus, Pituitary,** and **Pineal Gland**.
- **Peripheral Endocrine Glands:** **Thyroid, Parathyroid, Adrenal gland,** ovaries, testes, placenta and **Pancreas**.



General histological features of an Endocrine gland

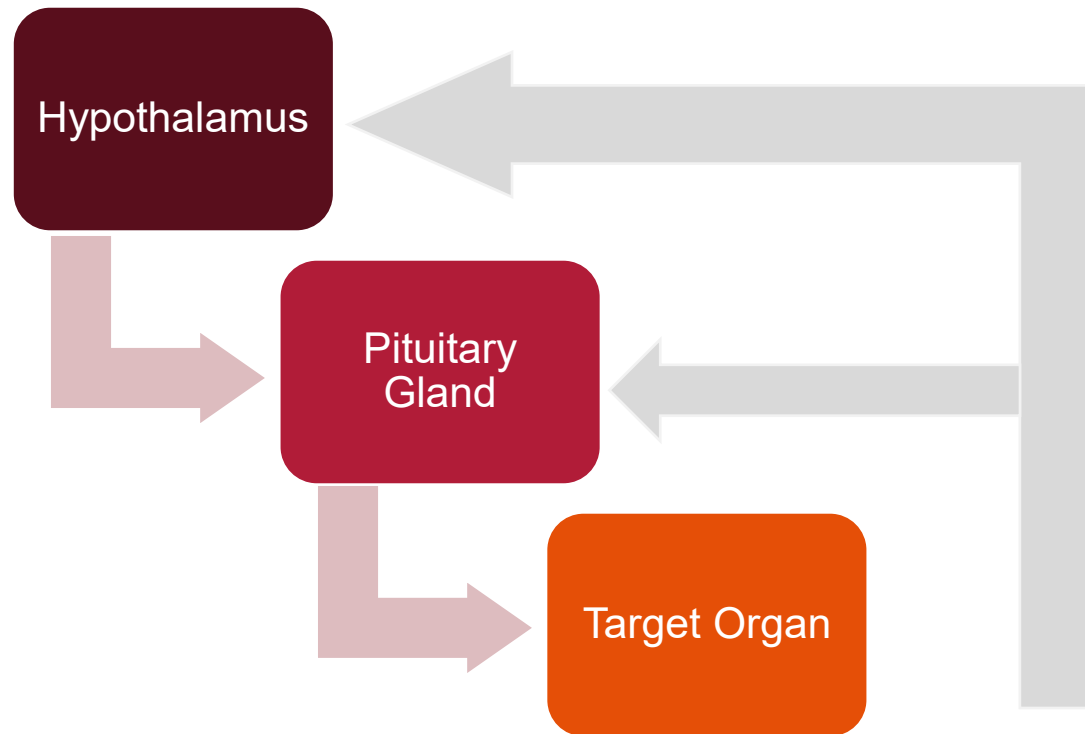
- **Lack** an **excretory duct**.
- Highly vascular with **fenestrated capillaries**.
- **Clusters** of hormone-producing cells.
- **Reticular/loose connective tissue** matrix
- **Encapsulated** organs.



High magnification light micrograph Mallory trichrome Stain Human Pituitary Gland Pars Distalis

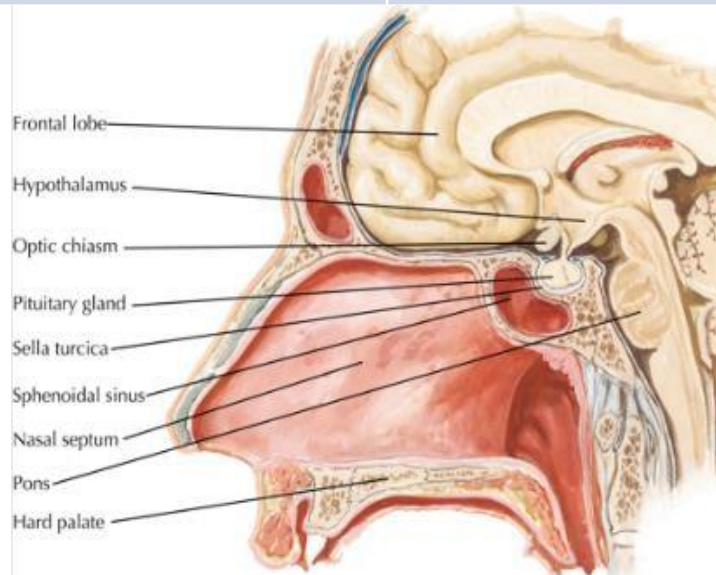
Endocrine Feedback System Blueprint

- Hypothalamo – Hypophyseal (Pituitary Gland) – Target Organ Axis
- **Most** endocrine feedback is **negative feedback**
- *to be continued in physiology...*

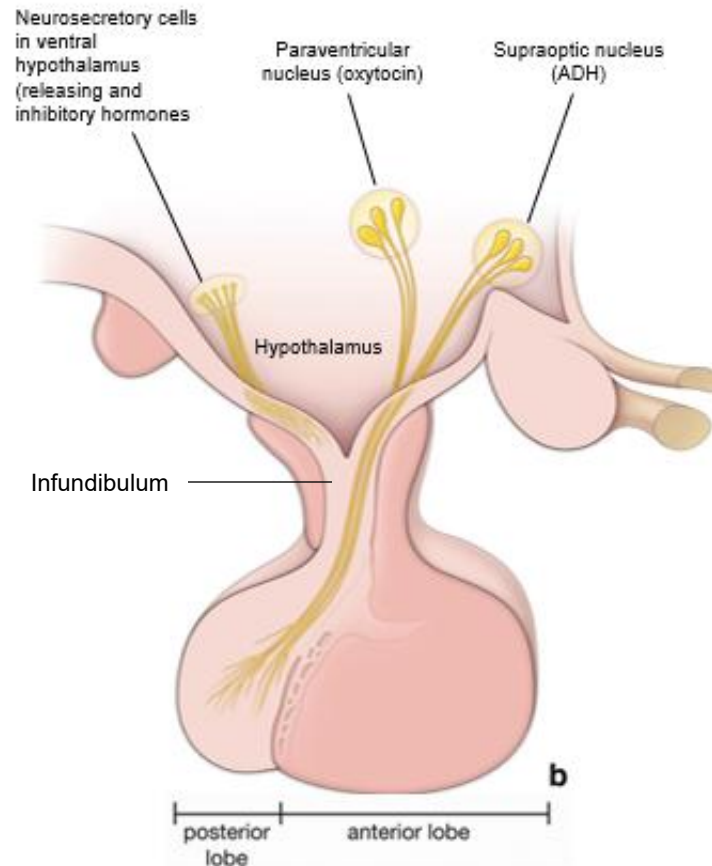


Hypothalamus and Pituitary Gland

Hypothalamus	Pituitary Gland
Anatomy: Located at the base of the brain surrounding the third ventricle. The inferior limit is the median eminence. Function: Major integrating link between the nervous and endocrine system. - Produces hormones that regulates the anterior pituitary gland activity. - Produces hormones that are stored in the posterior pituitary.	Anatomy: Located at the base of the brain, sitting in the sphenoid bone - Sella turcica. Inferior to the optic chiasm. Connected to the hypothalamus by the infundibulum. - Two distinct parts: Anterior Pituitary Gland (Adenohypophysis) Posterior Pituitary (Neurohypophysis) Blood supply: superior and inferior hypophyseal artery and hypophyseal portal system. Function: Produces hormones that regulates other endocrine organs. - Stimulated and inhibited by the hypothalamus and feedback mechanisms from target organs.

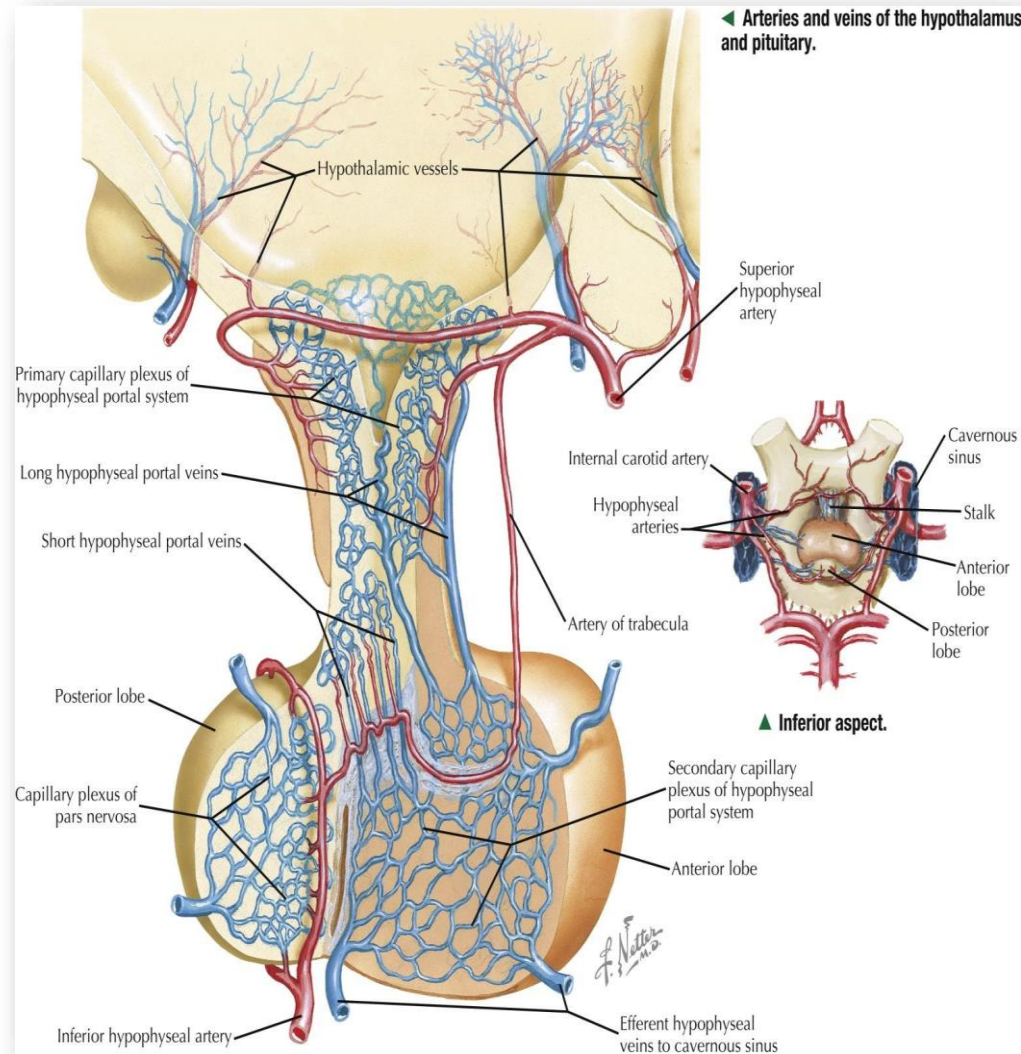


Hypothalamo-Hypophysis Relationships



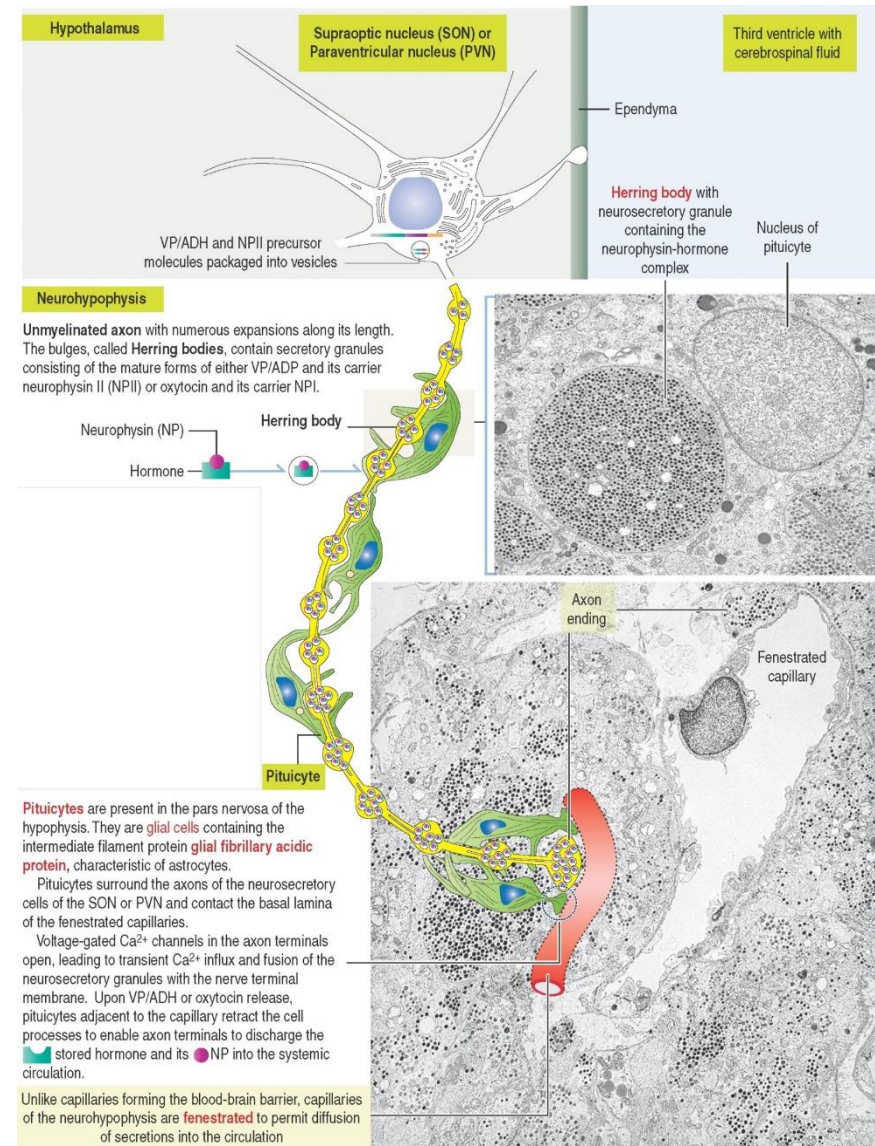
- **Parvocellular neurosecretory** neurons in ventral hypothalamus:
 - Synthesize and secrete stimulating and inhibitory hormones that regulate the **anterior pituitary gland** via the **hypothalamohypophyseal portal system**
 - Eg Growth Hormone Releasing Hormone (GHRH) or Growth Hormone Inhibiting Hormone (Somatostatin)
- **Magnocellular neurosecretory neurons:**
 - **Synthesize two hormones** that are transported, stored in and released from the **posterior pituitary gland** via the **hypothalamohypophyseal tract**
 - **ADH or Vasopressin** from Supraoptic Nucleus (SON)
 - Primarily acts on the kidneys to **increase H₂O reabsorption**
 - **Oxytocin** from Paraventricular Nucleus (PVN)
 - During parturition (birth) → **uterine contraction**
 - During lactation → **contraction of myoepithelial cells** of mammary gland

Hypothalamohypophyseal Portal System



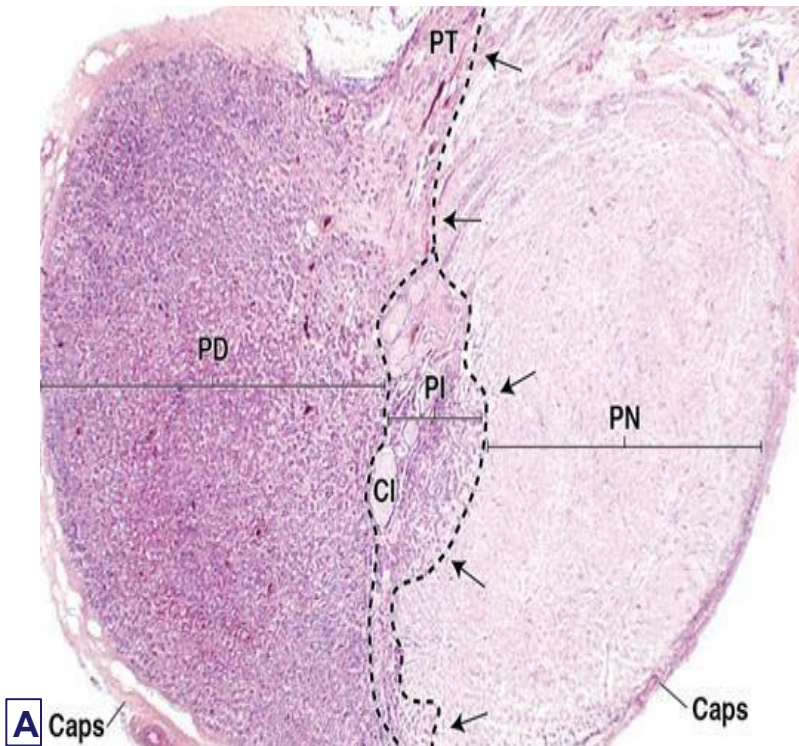
Hypothalamohypophyseal Tract

- **Unmyelinated axons** of PVN and SON
- Terminate on **fenestrated capillaries** in the neurohypophysis
- Tract passes through the infundibulum to the Pars nervosa
- Several dilations (**Herring Body**) containing **Hormones (ADH and Oxytocin)** bound to **neurophysin** (protective protein).
- Axons are protected/supported by **pituicytes**

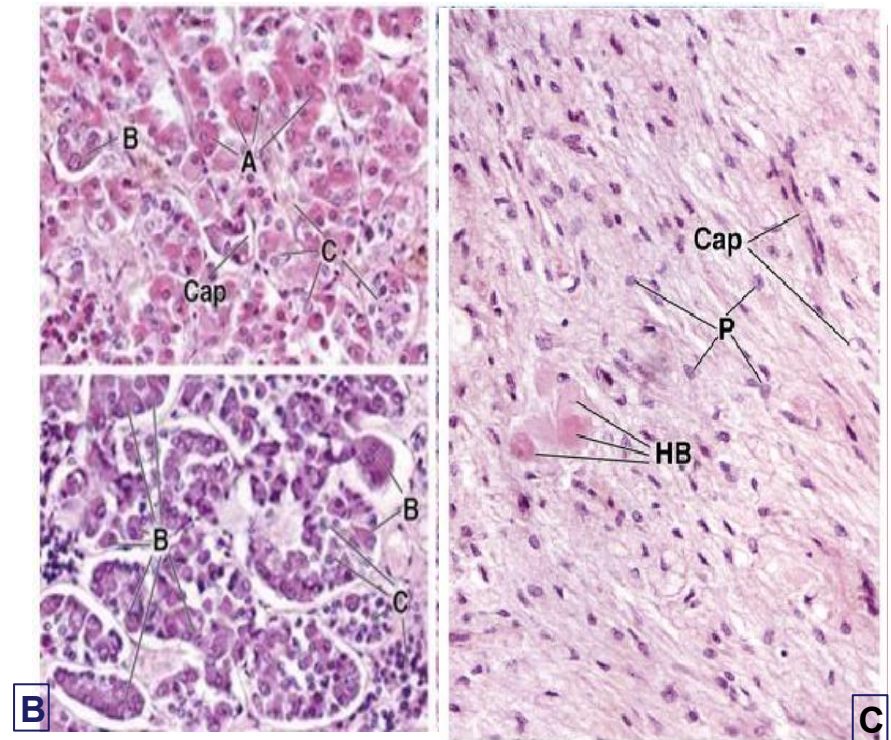


Pituitary Gland

- Anterior Pituitary: Granular appearance and **dark staining** on H&E
- Posterior Pituitary: **Pale staining** appearance on H&E



A. Light micrograph H&E stain of the human pituitary gland sagittal section

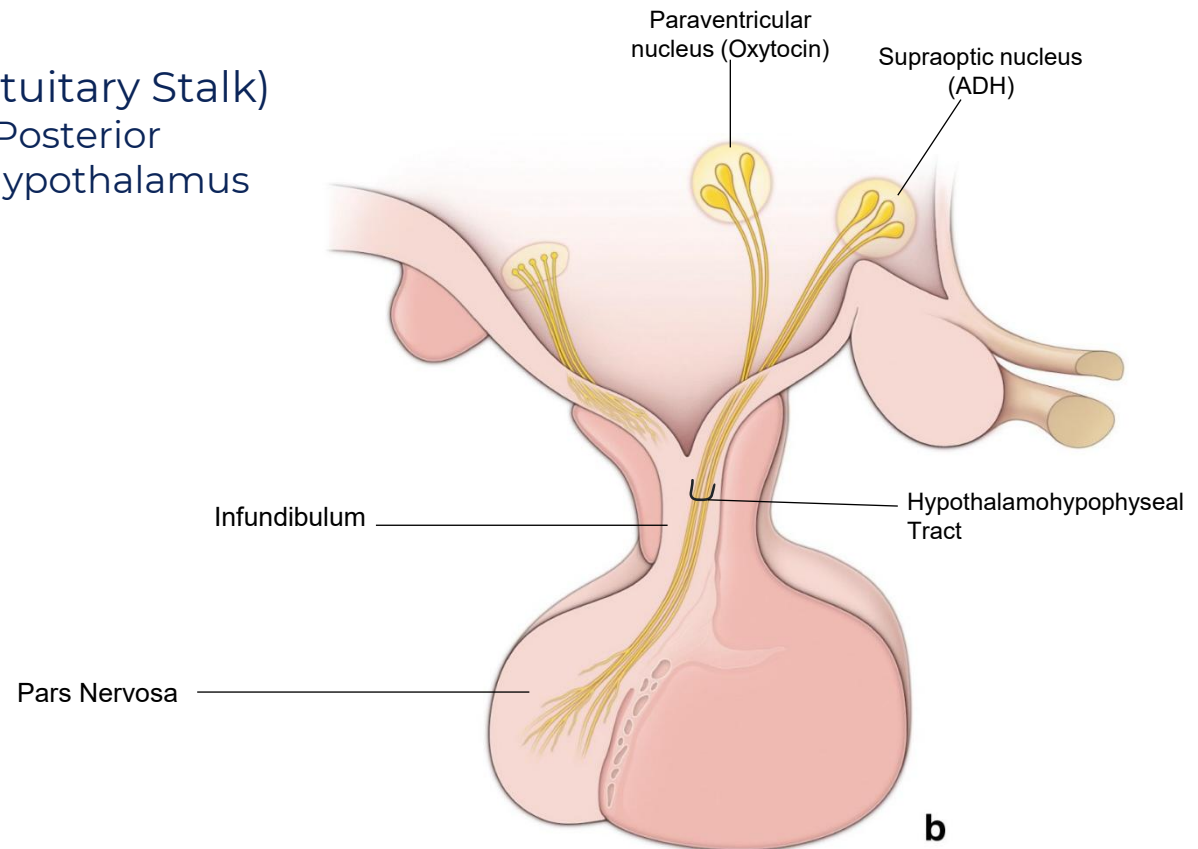


B. Medium magnification light micrograph H&E Stain of the human anterior pituitary gland

C. Medium magnification light micrograph GH&E Stain of the human posterior pituitary gland

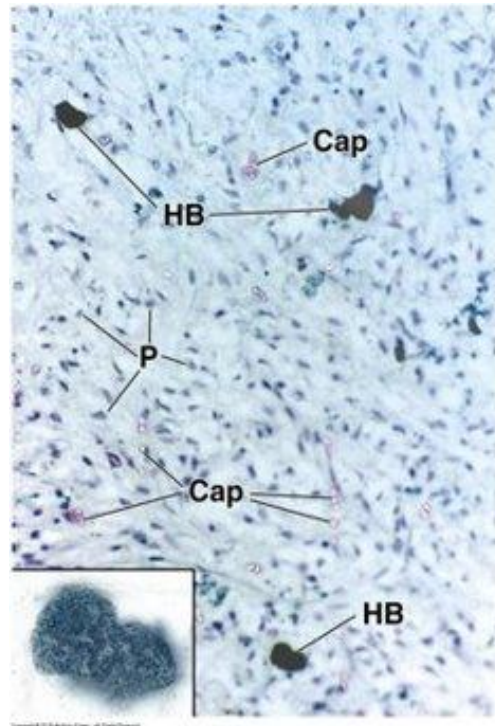
Posterior Pituitary Gland

- Storage and secretion of hormones synthesized in hypothalamic nuclei (SON and PVN)
- Posterior Pituitary or Neurohypophysis.
- Two Parts:
 - Pars Nervosa
 - Infundibulum (Pituitary Stalk)
 - Connects the Posterior Pituitary to the Hypothalamus

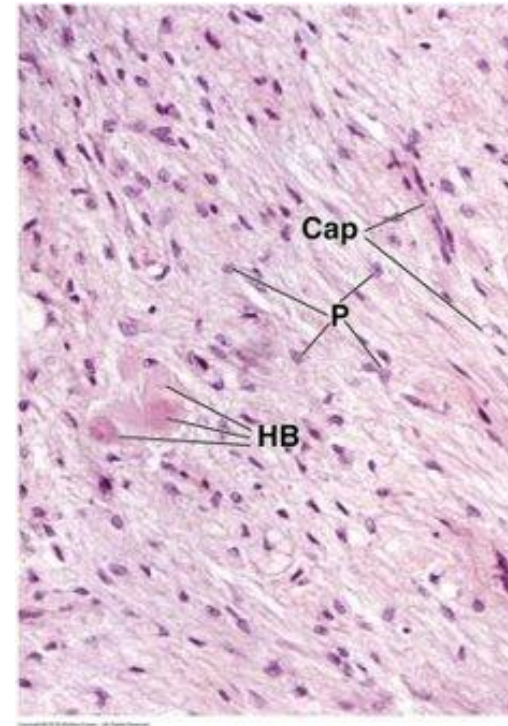


Posterior Pituitary Gland

- **Fibrotic in appearance because of the presence of axons.**
- **Herring bodies (HB):** dilation of unmyelinated axons containing hormones from the hypothalamus bound to a protective protein called **neurophysin**.
- **Pituicytes (P):** specialized glial cells **specific to the neurohypophysis** + Glial Fibrillary Acidic Protein (GFAP). Similar to astrocytes.
- **Fenestrated Capillaries (Cap)**



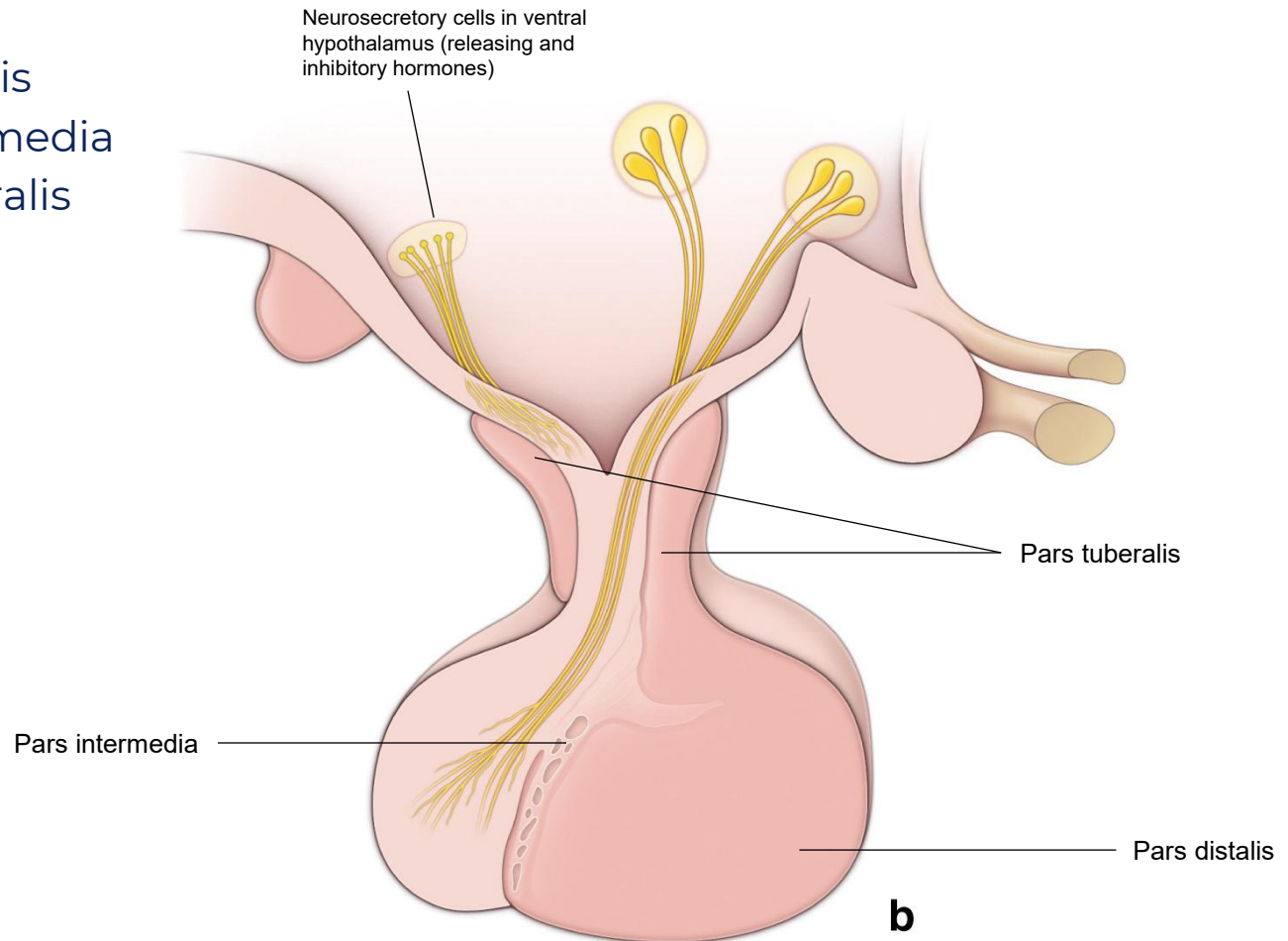
High magnification light micrograph Aniline Blue-Black stain human Pars nervosa (Posterior Pituitary Gland)



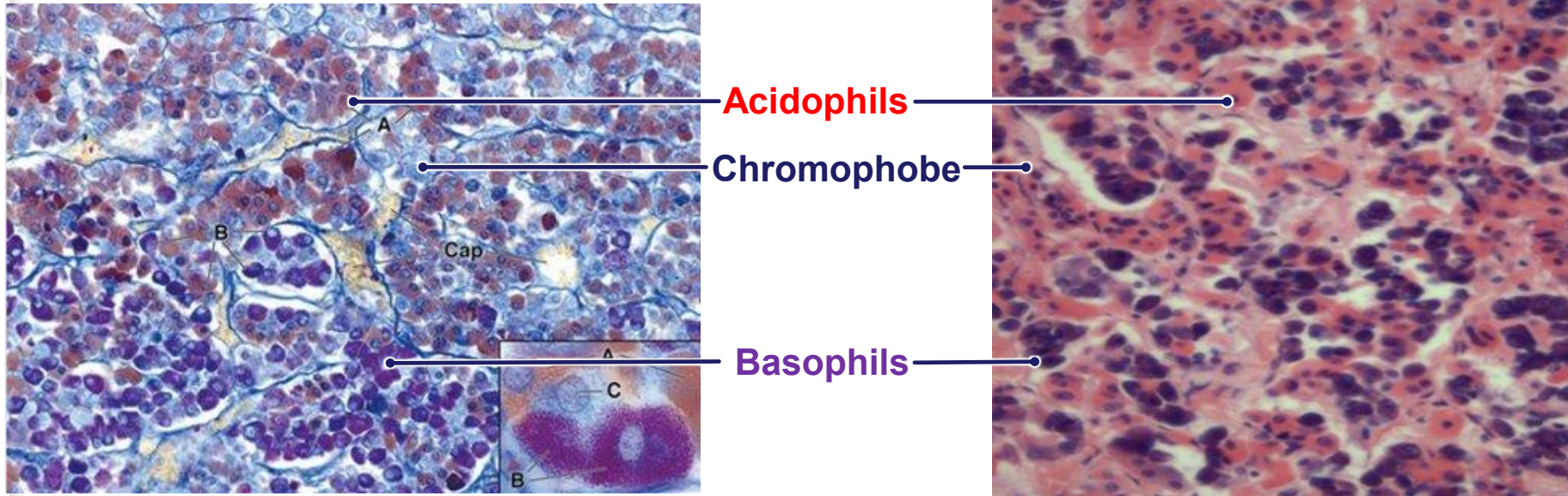
High magnification light micrograph H&E stain human pars nervosa (Posterior Pituitary Gland)

Anterior Pituitary Gland

- Master gland of the endocrine system *functionally* connected to the hypothalamus
- Anterior Pituitary or Adenohypophysis
- Three parts:
 - Pars Distalis
 - Pars Intermedia
 - Pars Tuberalis



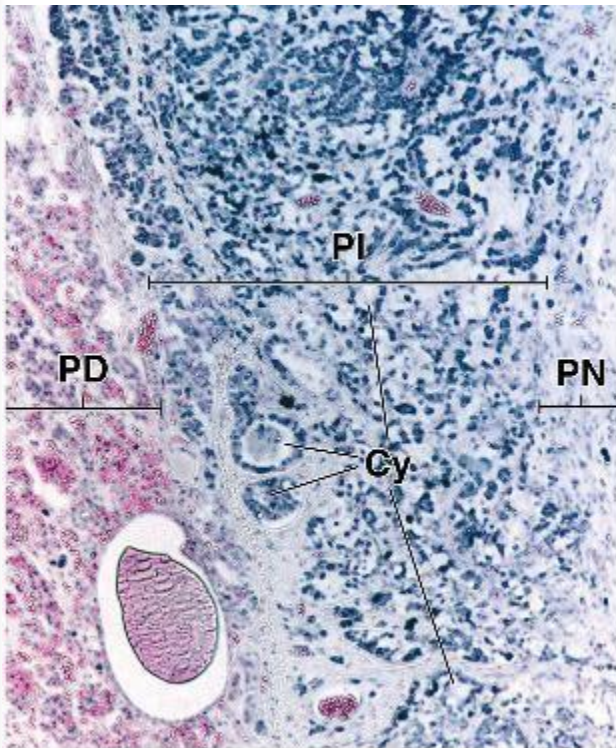
Adenohypophysis (Pars Distalis)



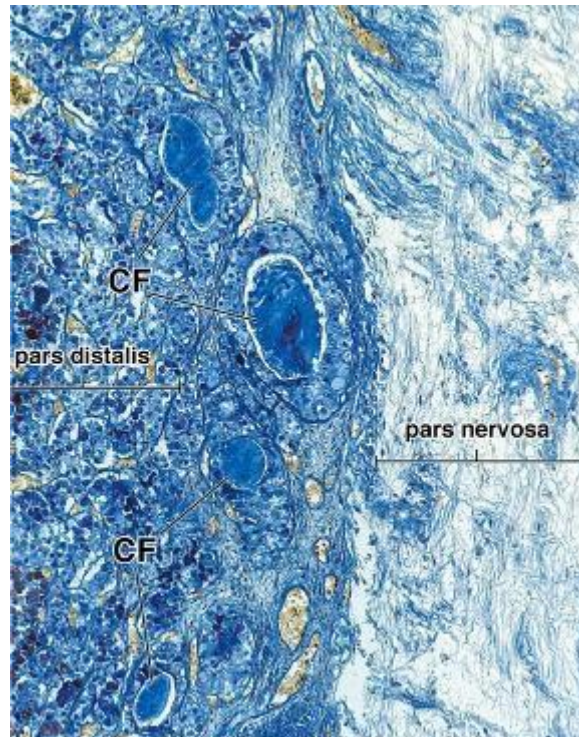
- Glandular appearance and general features of endocrine tissue
- Cell types are observed:
 - Chromophobes (50%) (C)
 - Chromophiles (50%)
 - **Acidophils (A)**
 - **Basophils (B)**
 - Folliculostellate cells

Adenohypophysis (Pars Intermedia)

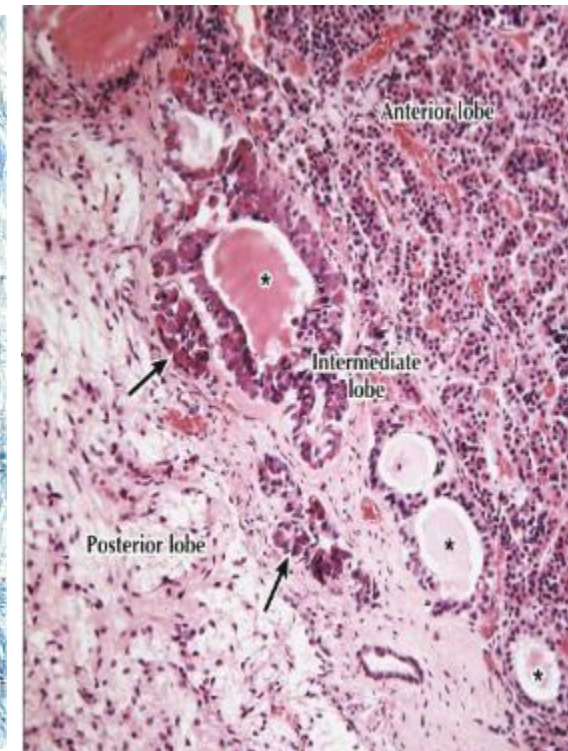
- **Rudimentary**
- **Small follicles** (or cysts [Cy]) filled with Colloid
- Cysts lined by **Basophilic cuboidal cells** assumed to be corticotropes
- Also contains **chromophobes**



Medium magnification light micrograph PAS/aniline blue-black Human Pituitary Gland



Medium magnification light micrograph Toluidine Blue stain Human Pituitary Gland Pars Intermedia with adjacent Pars Distalis and Pars Nervosa



Medium magnification light micrograph Hematoxylin & Eosin stain Human Pituitary Gland Pars Intermedia (colloid cyst present) with adjacent Pars Distalis and Pars Nervosa

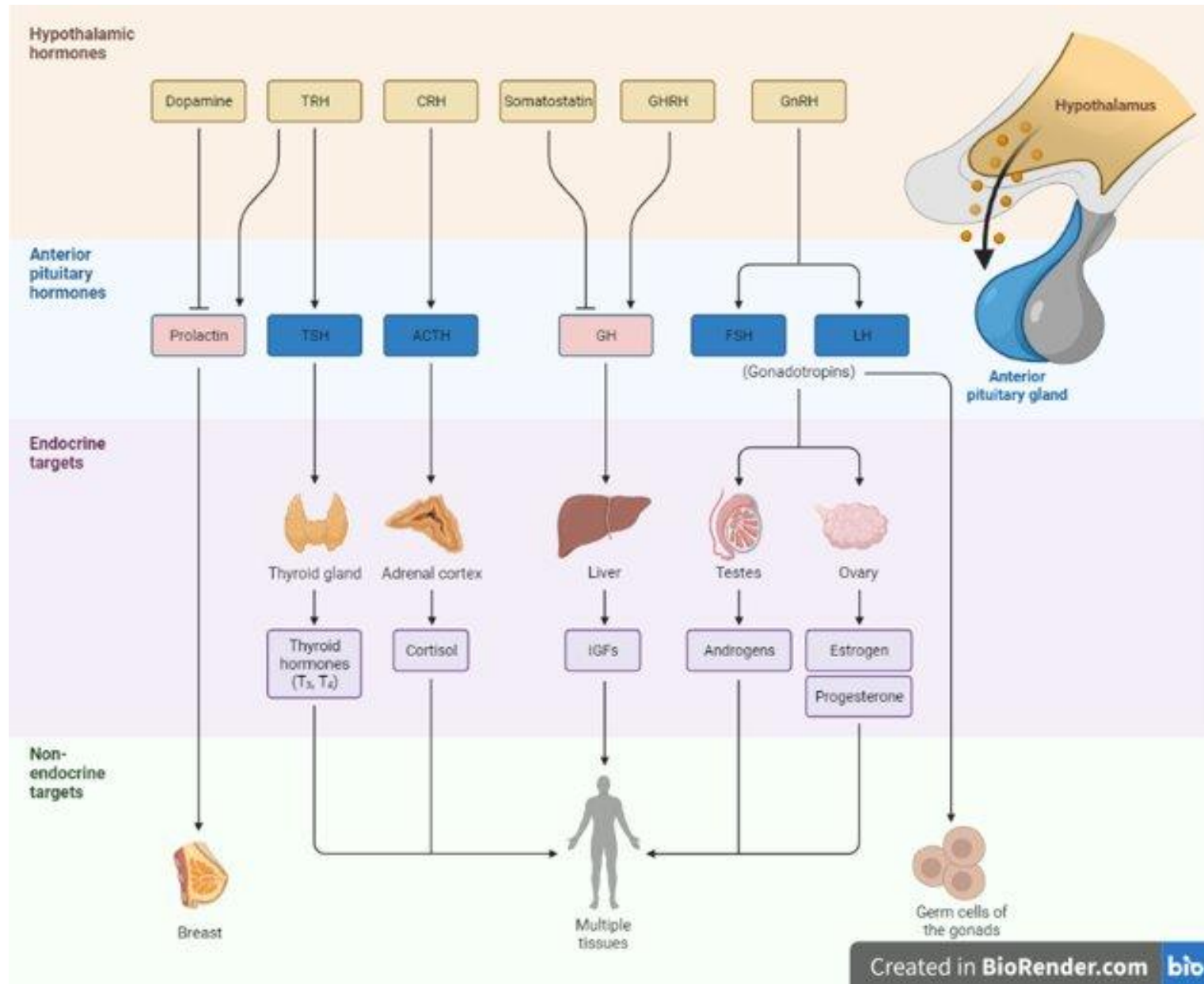
Anterior Pituitary Gland

Anterior Pituitary Gland Cell Types, Hormones, and Function

Cell Type	Hypothalamic Hormone Regulation	Percent of Cells in Gland	Hormone Secreted	Function
Somatotrope (GH Cell)	Stimulated by Growth Hormone Releasing Hormone (GHRH) Inhibited by Somatostatin	50	Growth Hormone (GH)	Stimulates the liver and other organs to synthesize and secrete insulin-like growth factor I (IGF-I). IGF-1 acts on the growth plates of bone and skeletal muscle resulting in growth
Lactotrope (PRL Cell)	Stimulated by Thyrotropin Releasing Hormone (TRH) Inhibited by Dopamine	15 – 20	Prolactin (PRL)	Promotes milk secretion from mammary glands
Corticotrope (ACTH Cell)	Stimulated by Corticotrophin Releasing Hormone (CRH) Inhibited by elevated levels of cortisol.	15 – 20	Proopiomelanocortin (POMC) → Adrenocorticotrophic hormone (ACTH) and B- lipotropic hormone (B-LPH), Melanocyte Stimulating Hormone (MSH), Beta Endorphins, and Enkephalin	ACTH stimulates hormone secretion from the adrenal cortex
Gonadotrope (FSH and LH Cells)	Stimulated by Gonadotropin-Releasing Hormone (GnRH) Inhibited by elevated levels of estrogen or testosterone. **can be inhibited by PRL from lactotrophs	10	Follicle Stimulating Hormone (FSH) and Luteinizing Hormone (LH)	In females FSH acts on the ovary to promote development of ovarian follicles. In males FSH stimulates spermatogenesis. In females LH stimulates maturation of ovarian follicles, ovulation, and corpus luteum formation. In males stimulates Leydig cells to produce androgens
Thyrotrope (TSH Cell)	Stimulated by Thyrotropin Releasing Hormone (TRH) Inhibited by Somatostatin	~5	Thyroid Stimulating Hormone	Stimulates production and release of thyroglobulin and thyroid hormones

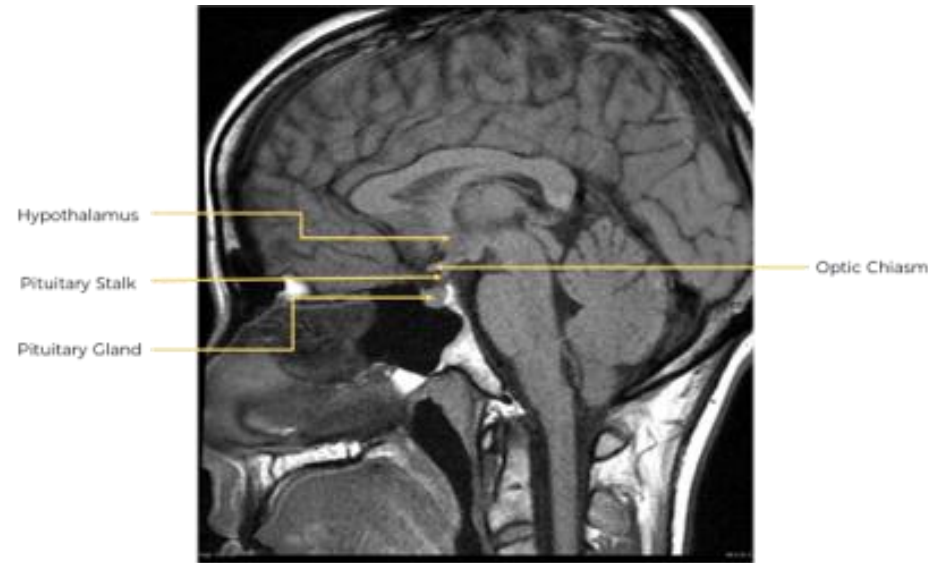
Chromophile Type	Specific Cell Types
Acidophils	Somatotrophs Mammotrophs / Lactotrophs
Basophil	Gonadotrope Corticotrope Thyrotrope

Adenohypophysis Hormones



Pituitary Adenoma

- Most are slow-growing and benign
- 10 – 15% of all brain tumors
- Classified based on **size** or **cell of origin**
- Maybe functioning (secrete hormones) or nonfunctioning (do not secrete hormones)
- General symptoms
 - **Headaches**
 - **Vision problems** (double vision, vision loss)
 - **bitemporal hemianopsia**)
 - Nausea or vomiting
 - Changes in behavior, including hostility, depression and anxiety
 - Nasal drainage
 - Nonfunctioning tumors present with symptoms of mass effect
 - **Hypopituitarism**



From Fernandes V, Santos MJ, Almeida R, Marques O. Ten-year follow-up of a giant prolactinoma. BMJ Case Rep. 2015 Nov 20;2015:bcr2015212221. doi: 10.1136/bcr-2015-212221. PMID: 26590188; PMCID: PMC4680285.

Pituitary Adenoma

FULL TEXT ARTICLE

Pituitary Tumor With Gigantism, Acromegaly and Preclinical Cushing's Disease Diagnosed from the 10th Row



John B. Tourtelot MD and David L. Vesely MD, PhD

American Journal of the Medical Sciences, The, 2013-08-01, Volume 345, Issue 2, Pages 169-171, Copyright © 2013 Southern Society for Clinical Investigation

Chromophille Type	Speciflc Cell	Adenoma	Features
Acidophils	Somatotropes	Somatotropic Adenoma Growth Hormone Adenoma	Adults → Acromegaly Adolescences → Gigantism
	Mammotrope / Lactotrophs	Prolactinoma	Nipple Secretion Amenorrhea (loss/absence of menstrual cycle) Sexual dysfunction
Basophils	Corticotrope	Corticotropic Adenoma	Cushing's Disease (e.g., weight gain, truncal obesity, purple striae-stretch marks on abdomen, chest, and thighs)
	Thyrotrope	Thyrotropic Adenoma	Hyperthyroidism (e.g., weight loss, tachycardia, increase systolic BP, muscle wasting, exophthalmos, goiter)

Pineal Gland

• Anatomy

- Posterior wall of 3rd ventricle

• Histological features

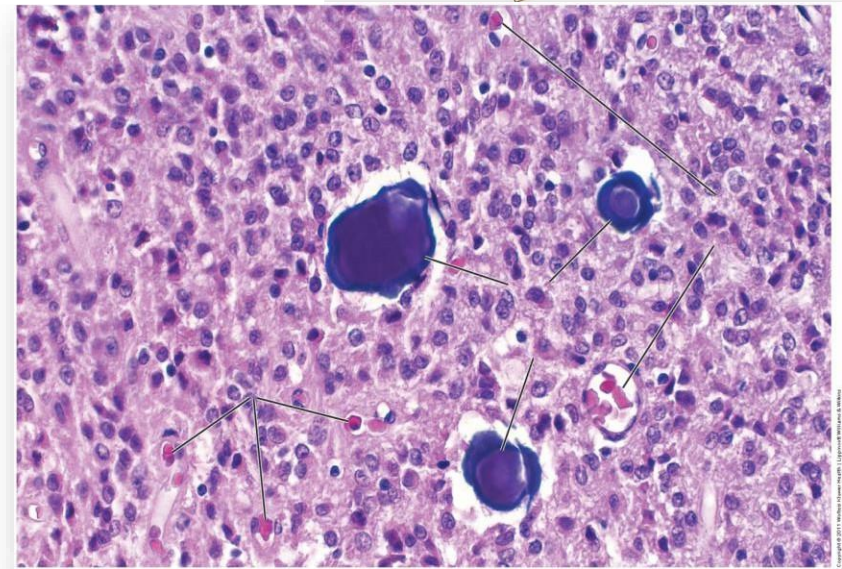
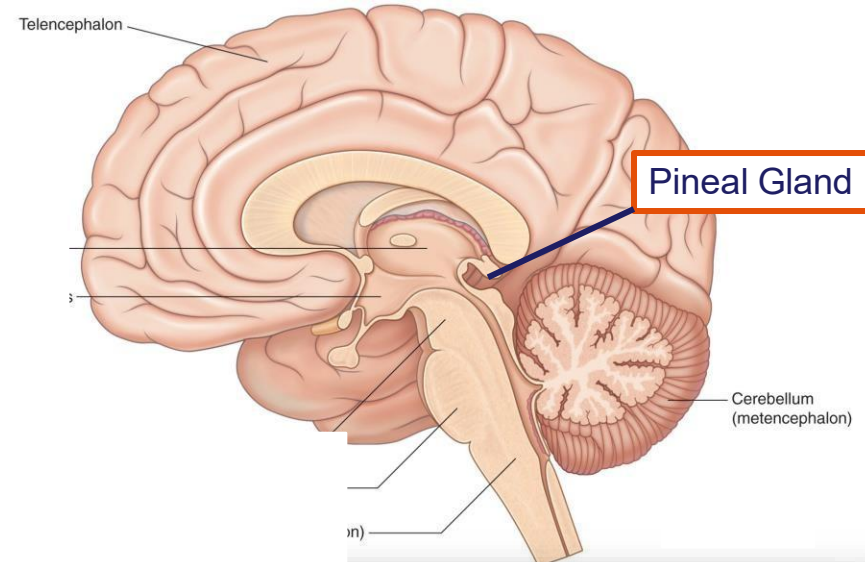
- Pinealocytes arranged in cords
- Interstitial Glial cells similar to Astrocytes
- Corpora Arenacea "Brain Sand"
- Calcifications increase over time
- Radiological Landmark

• Function

- Produces melatonin a hormone involved in the sleep wake cycle/ circadian rhythm
- **Melatonin can also inhibit GnRH production from the hypothalamus**

• Embryology

- Evagination from diencephalon at the roof of the third ventricle



High magnification light micrograph H&E stain Human Pineal Gland

Thyroid Gland

• Anatomy

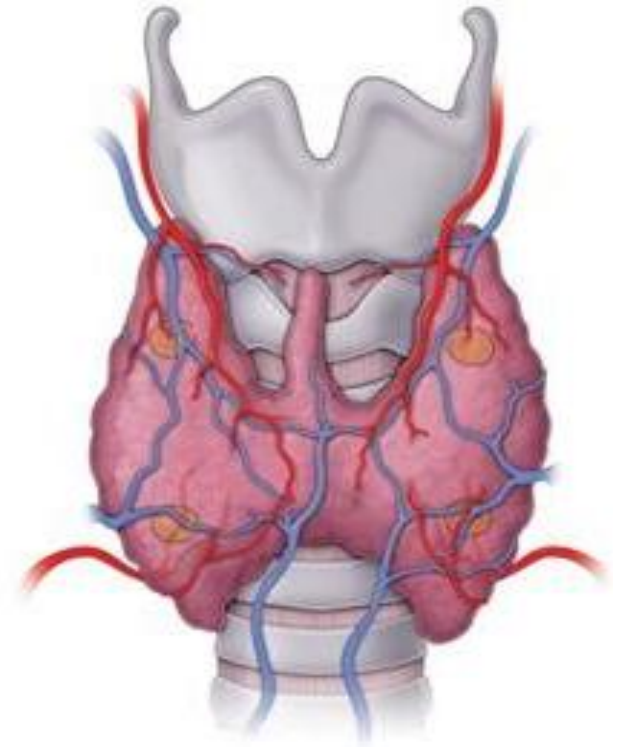
- Butterfly shaped organ in the anterior neck region
- Two lobes connected by an isthmus
- Superior thyroid artery arising from the external carotid artery, and the inferior thyroid artery branching from the thyrocervical trunk.

• Function

- Produces thyroid hormones: **Triiodothyronine (T3)** and **Thyroxine (T4)**: regulates metabolic activity.
- Produces **calcitonin**: lowers blood calcium levels by suppressing osteoclast activity (inhibiting bone resorption and stimulating absorption of calcium by the bones.)

• Embryology

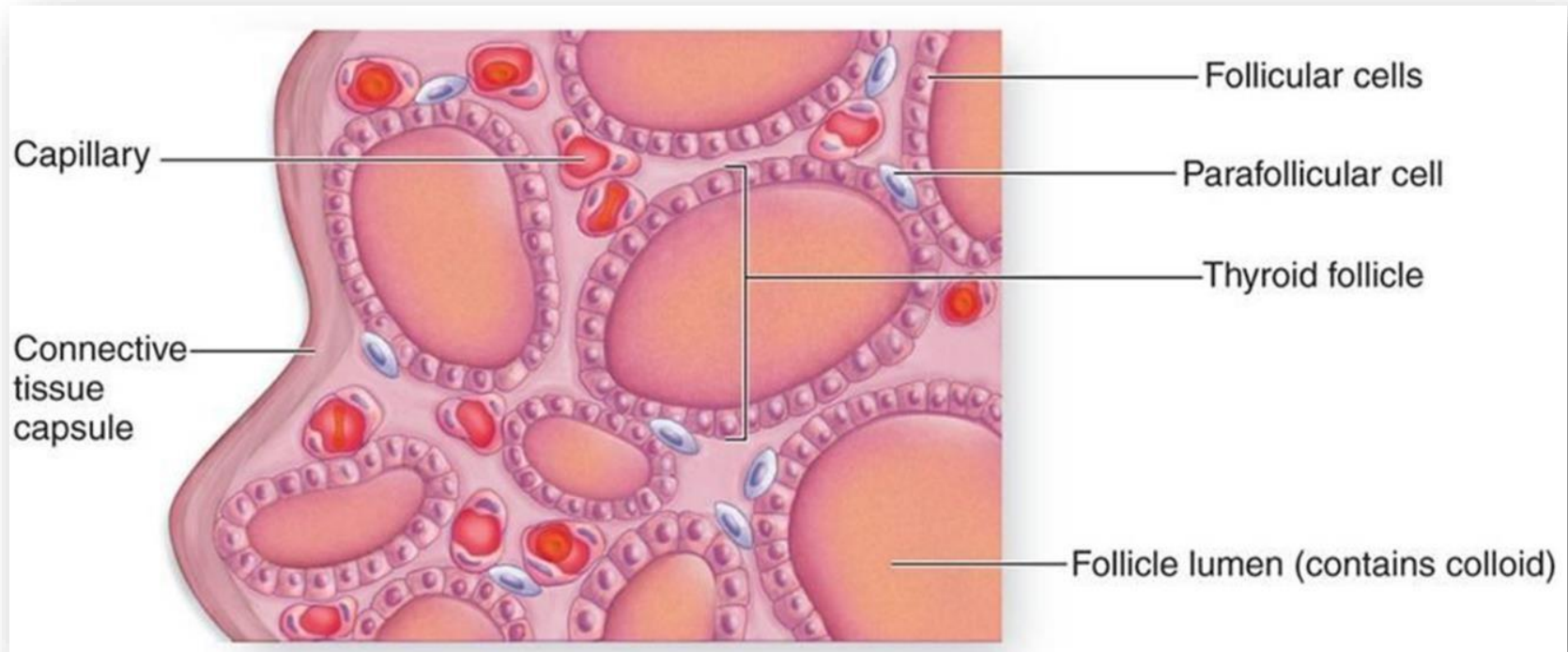
- Down growth from the base of the tongue thyroglossal duct (from endoderm), invasion of neural crest cells to form parafollicular cells.



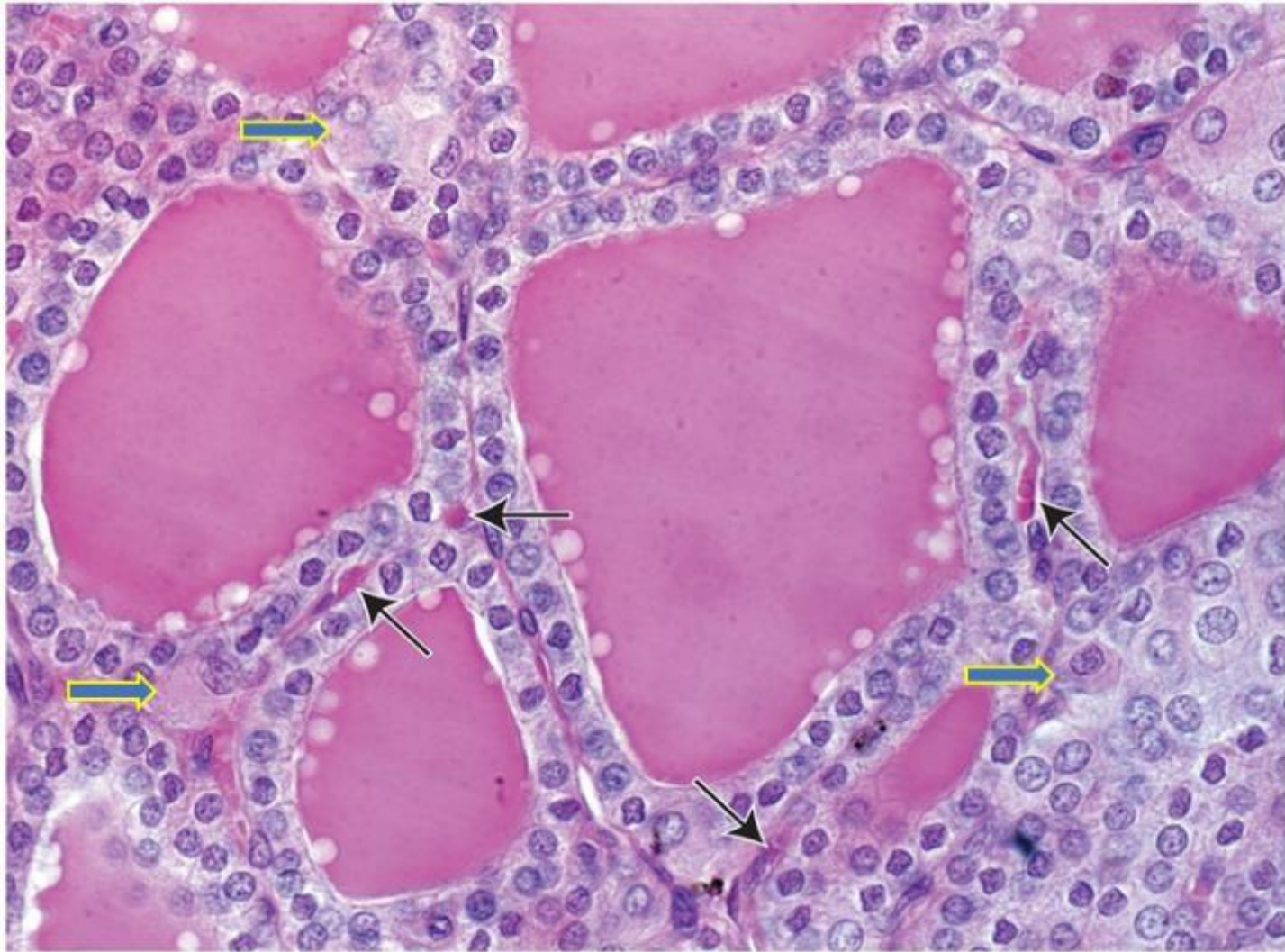
 indicates locations of parathyroid glands on the posterior surface of the lateral lobes

Thyroid Gland

- **Follicles** – site of **Thyroid hormones** synthesis
 - Colloid filled spaces lined by simple cuboidal epithelium with microvilli called **follicular cells (stimulated by TSH to produce Thyroglobulin and T3 + T4)**
 - Follicular cells possess features of protein secreting cells on electron microscopy.
- **Parafollicular cells (Clear Cells or C-Cells)**
 - Large polyhedral solitary or clustered within basal lamina of follicles.
 - Synthesize and **secrete calcitonin**: decreases blood calcium levels.



Thyroid Gland



High magnification light micrograph H&E Stain Human Thyroid gland. **Black Arrows point to capillaries. Yellow Highlighted arrows to Parafollicular cells**

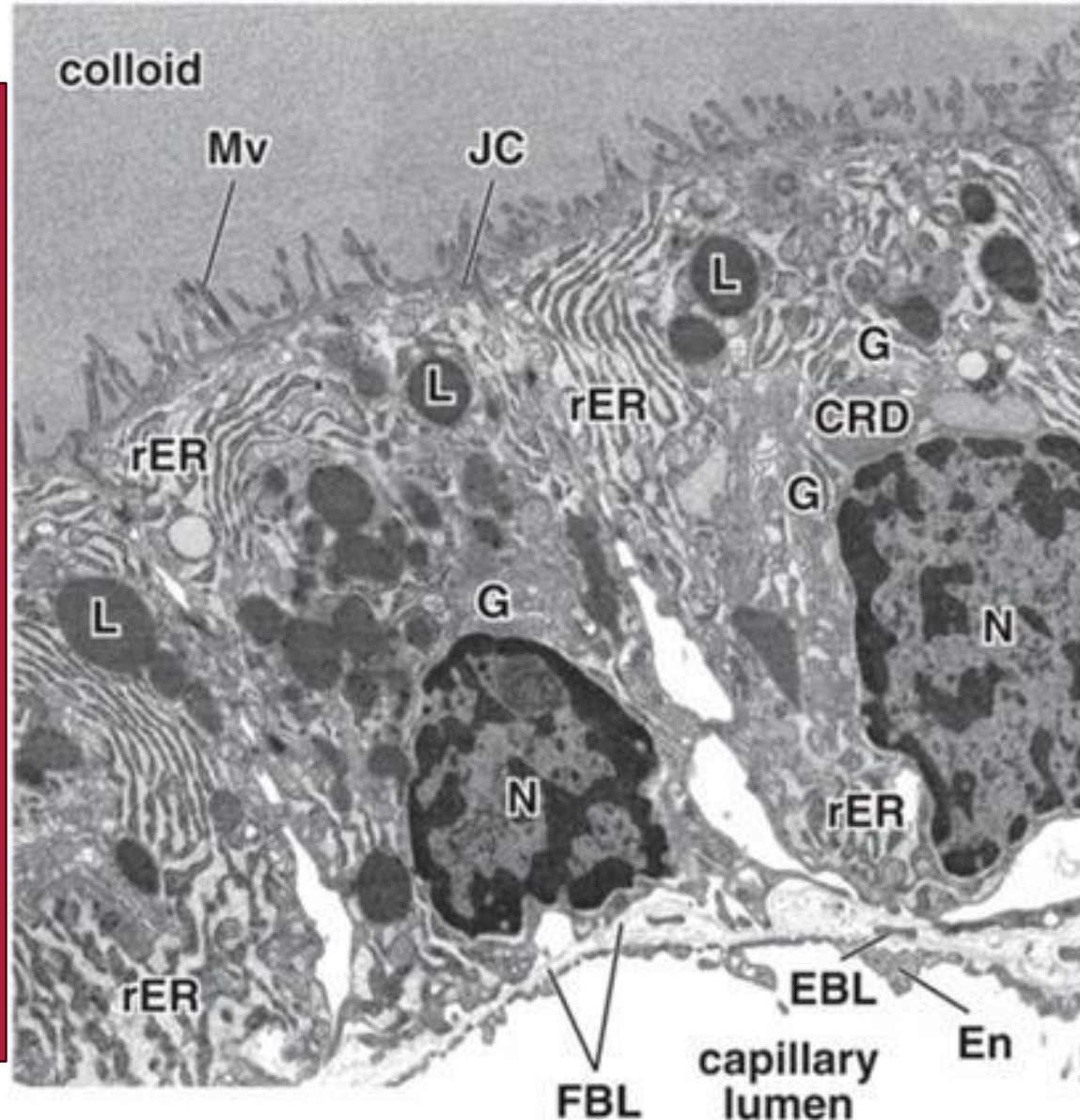
Thyroid Gland

Apical part of follicular cell
in contact with colloid.
Location of IODIDE/
CHLORIDE Transporter

Basal part of follicular cell
in contact with basal lamina
and blood vessels. Location
of SODIUM / IODIDE
SYMPORTER

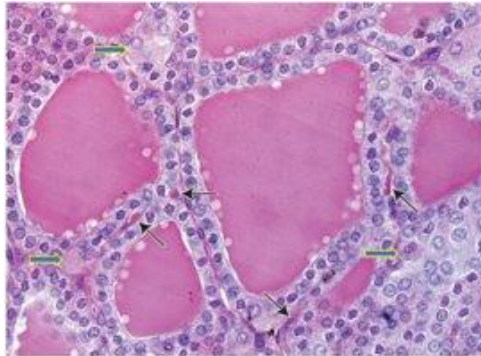
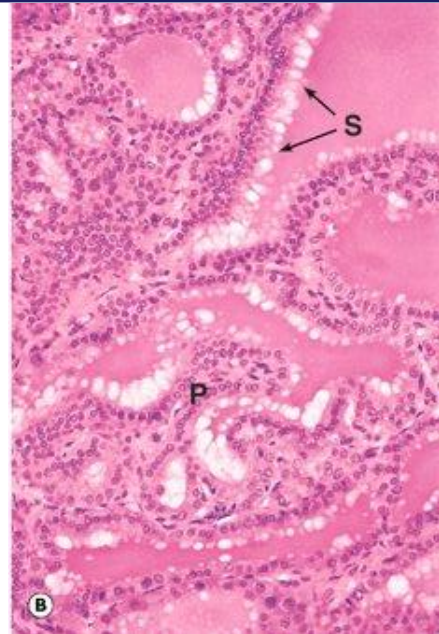
Ultrastructure of follicular
cells linked to their function.

- Abundant rER for thyroid hormone synthesis. Also accounts for basophilia of the follicular cell's cytoplasm on light microscopy
- Microvilli for absorption of substances present in colloid
- Euchromatic nucleus



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Thyroid Epithelium Activity

Decreased stimulation of follicular cells → decrease in the size of follicular cells to the cuboidal → squamous (flat) cells with abundant colloid in cavity	Normal Activity → cuboidal cells	Overstimulation the follicles are highly active → follicular cells increase in height → simple columnar with scanty colloid
 Simple Squamous	 Simple Cuboidal	 Simple Columnar
		

Parathyroid Gland

• Anatomy

- Two paired glands located posterior to the thyroid gland along the upper and lower poles of the thyroid gland lobes.
- Embedded within the connective tissue of the thyroid

• Function

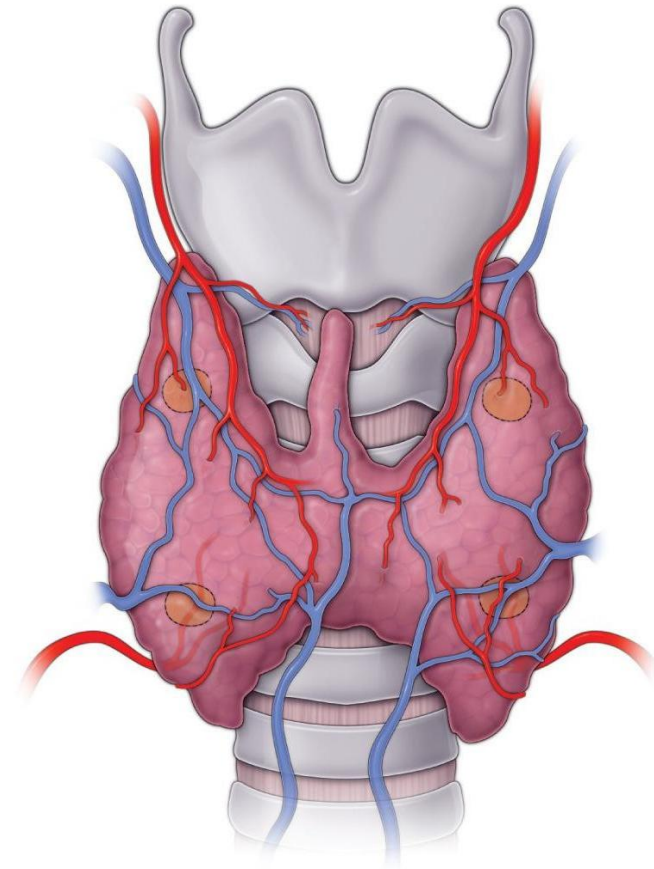
- Regulates blood calcium levels (increases calcium while decreasing phosphate levels)

• Histology

- Two cell Types present
 - Principal Cells or Chief Cells
 - Oxyphil Cells

• Embryology

- Derived from the 3rd and 4th pharyngeal pouch



indicates locations of parathyroid glands on the posterior surface of the lateral lobes

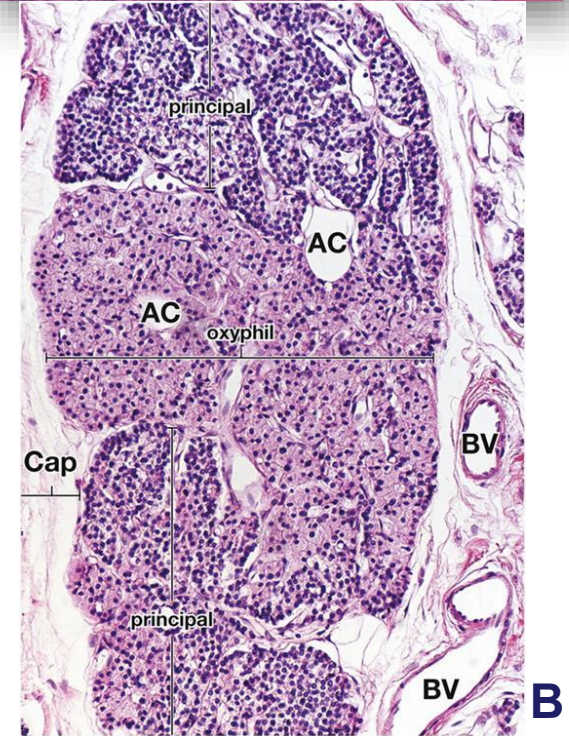
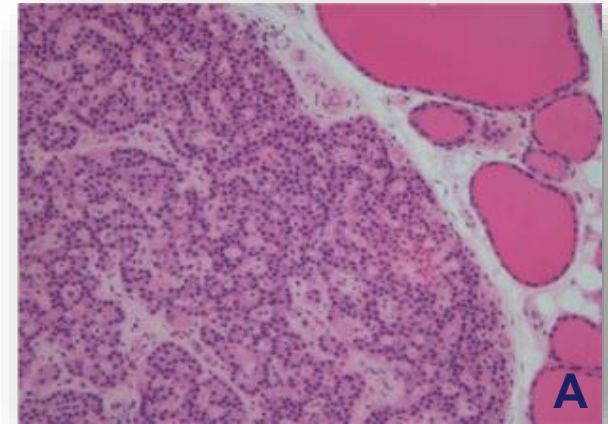
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Parathyroid Gland

- Embedded within connective tissue posterior to thyroid gland

Two cell types

- (A) Principal or Chief cells**
 - Small densely packed
 - Produce Parathyroid Hormone (PTH).
 - Increase blood calcium**
 - Increase renal tubular reabsorption of calcium
 - Increase activated vitamin D
 - Indirectly** increases Osteoclastic activity
 - Decrease blood phosphate levels**
 - Increase urinary excretion of phosphate
 - Think of PTH as the **Phosphate Trashing Hormone**
- (B) Oxyphil cells**
 - Large acidophilic cells
 - Function - unknown

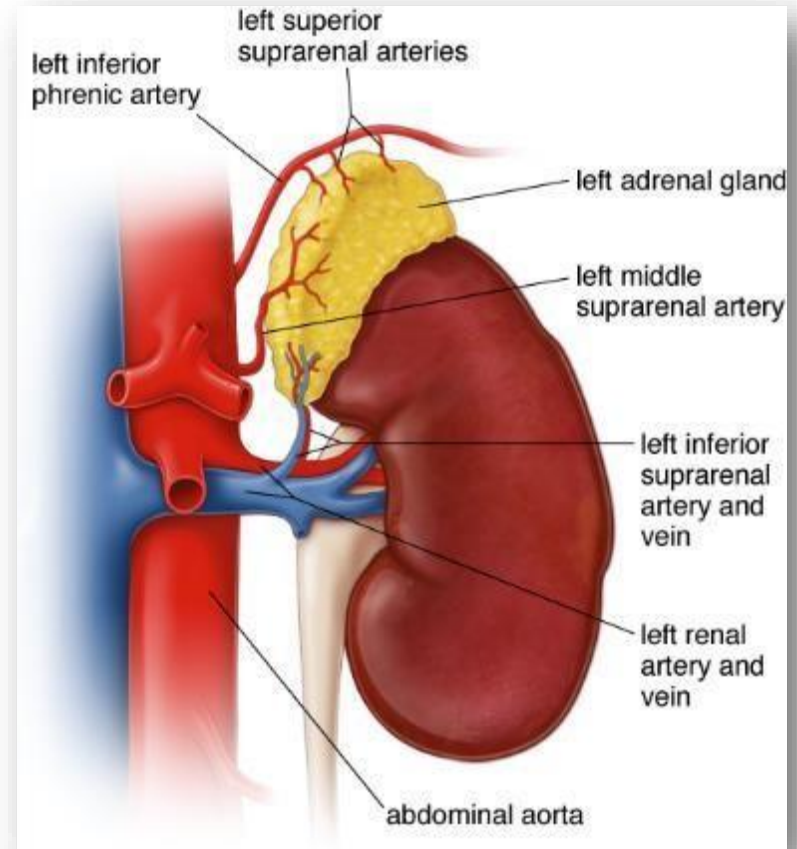


A: Low magnification H&E light micrograph parathyroid and thyroid gland
B: Low magnification H&E light micrograph parathyroid gland

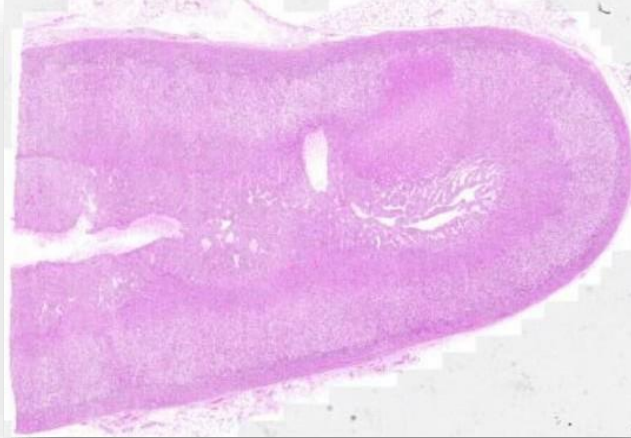
Adrenal Glands

AKA Suprarenal Glands. Anatomy

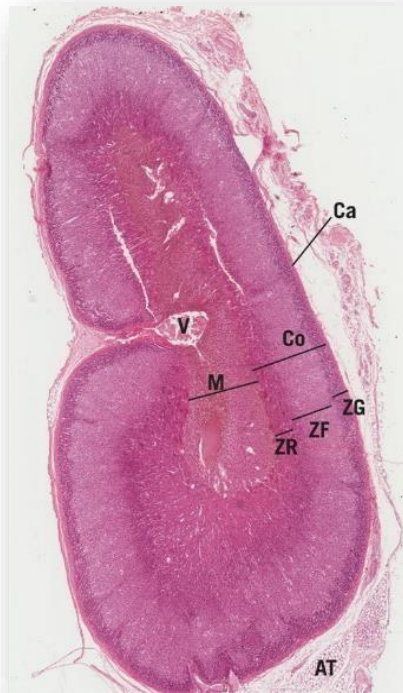
- Retroperitoneal organs
- Located along the superior pole of the kidneys
- Two distinct parts:
 - Cortex
 - Medulla
- **Function**
 - **Cortex** produces **Aldosterone, Cortisol,** and **weak Androgen hormones**
 - **Medulla** synthesizes and secrete **epinephrine and norepinephrine** key hormones of the sympathetic nervous system.
- **Embryology**
 - Dual Embryonic Origins
 - Cortex from mesenchyme
 - Medulla from neural crest cells



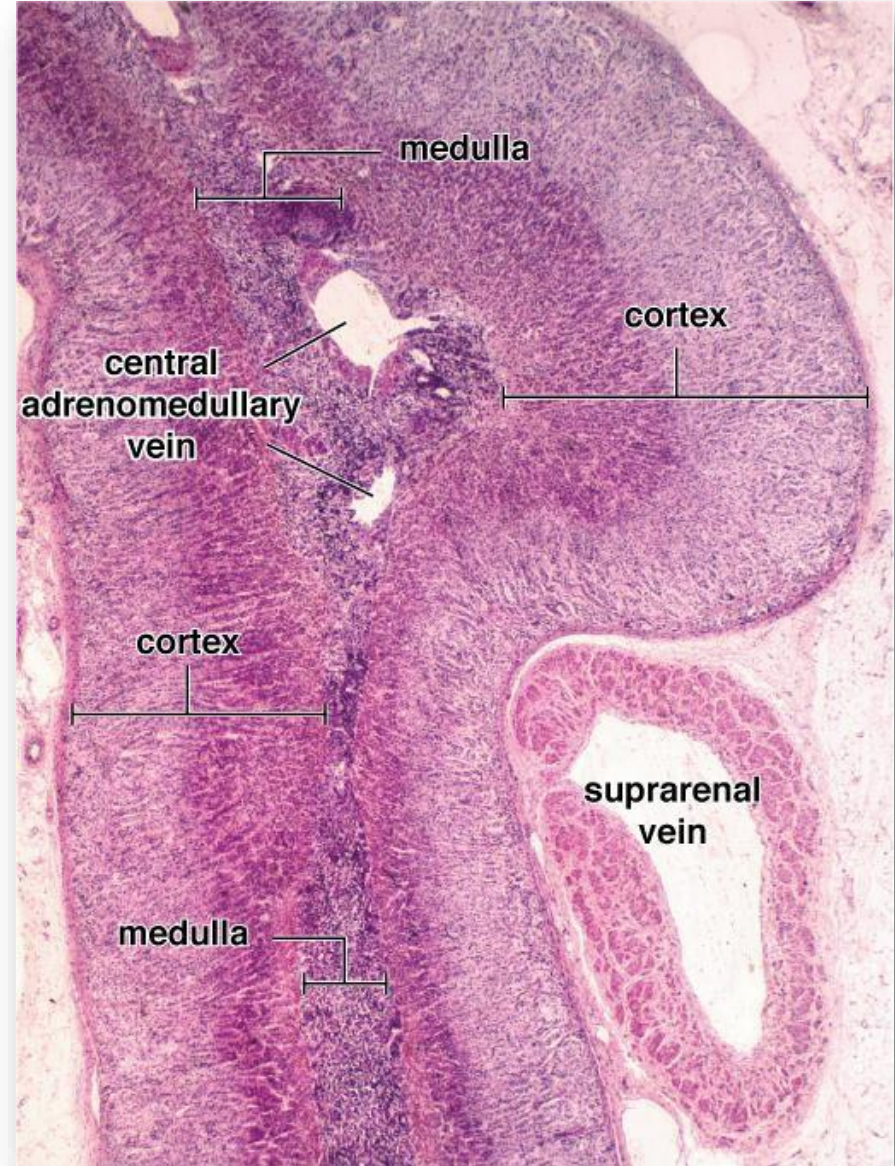
Adrenal Glands



Low Magnification H&E-stained Adrenal Gland with cortex and medulla present

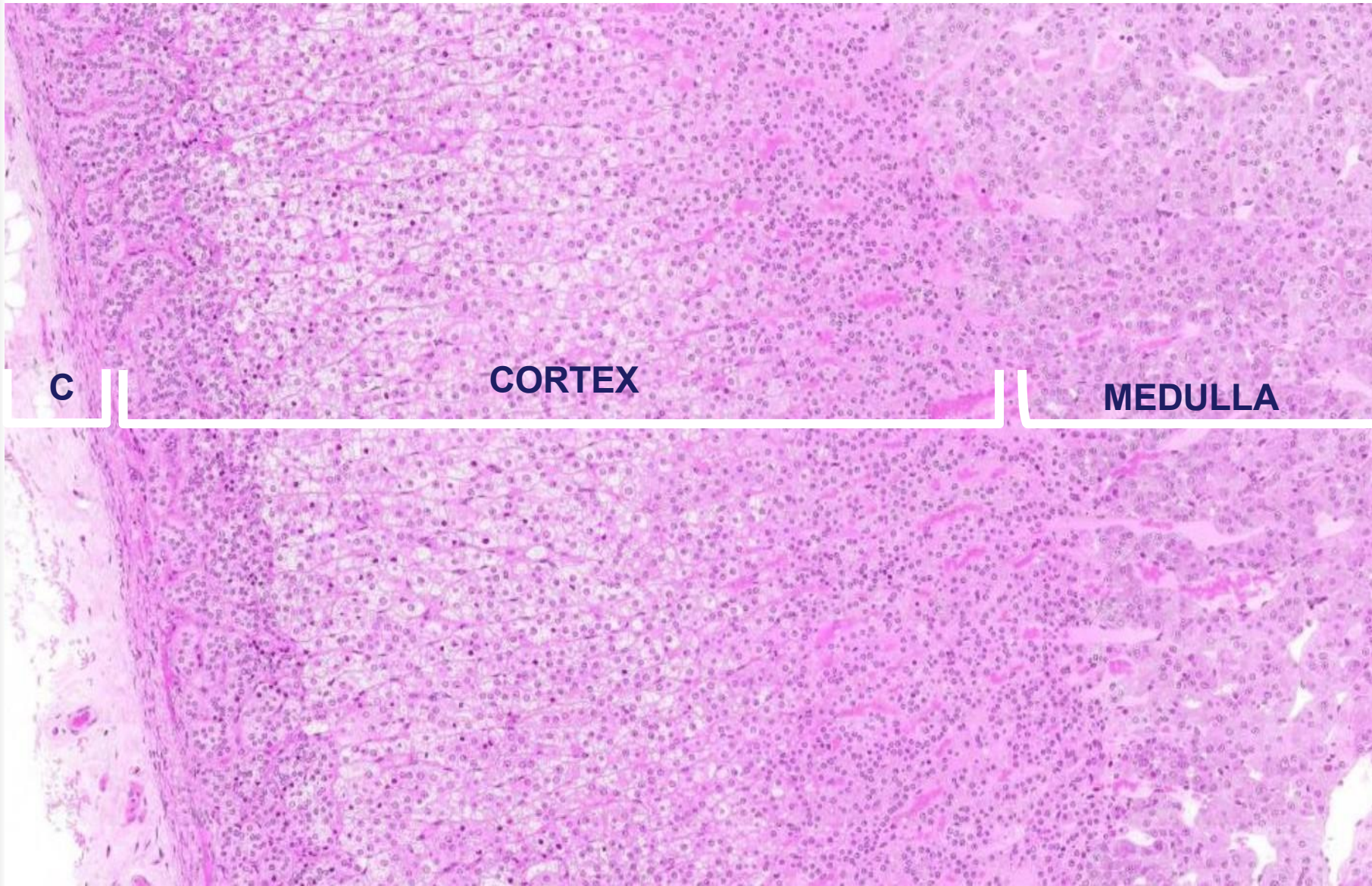


Low Magnification H&E-stained Adrenal Gland with cortex and medulla present



Low Magnification H&E-stained Adrenal Gland with cortex and medulla present along with suprarenal vein

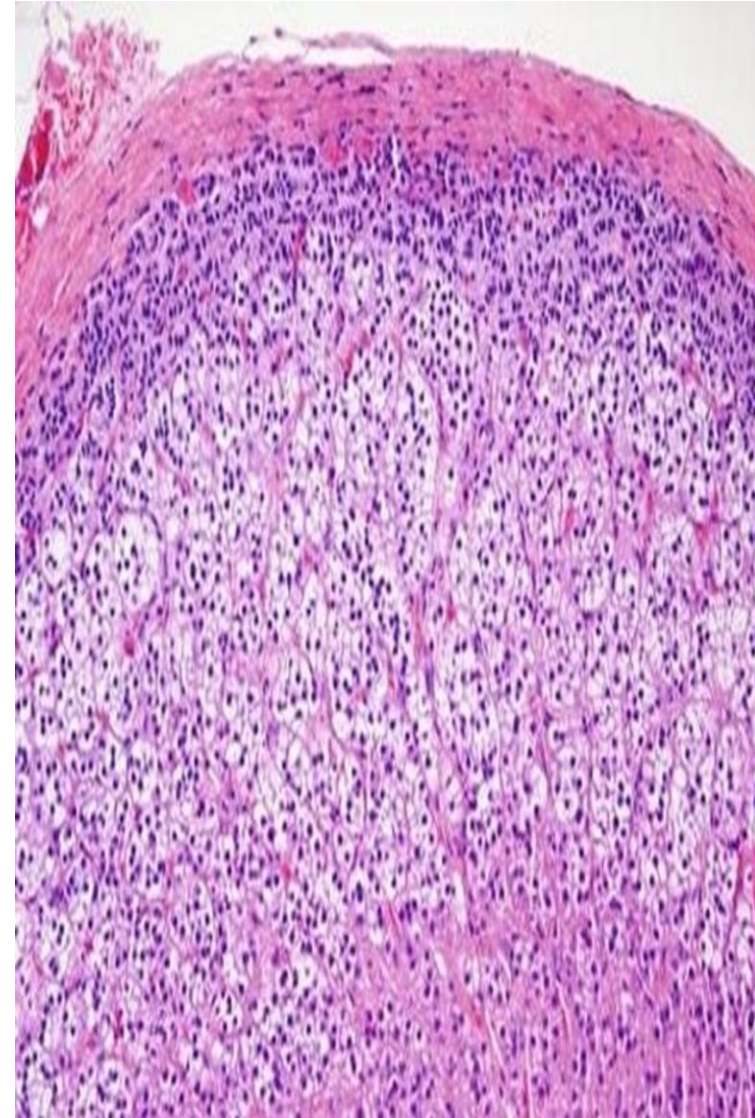
Adrenal Glands

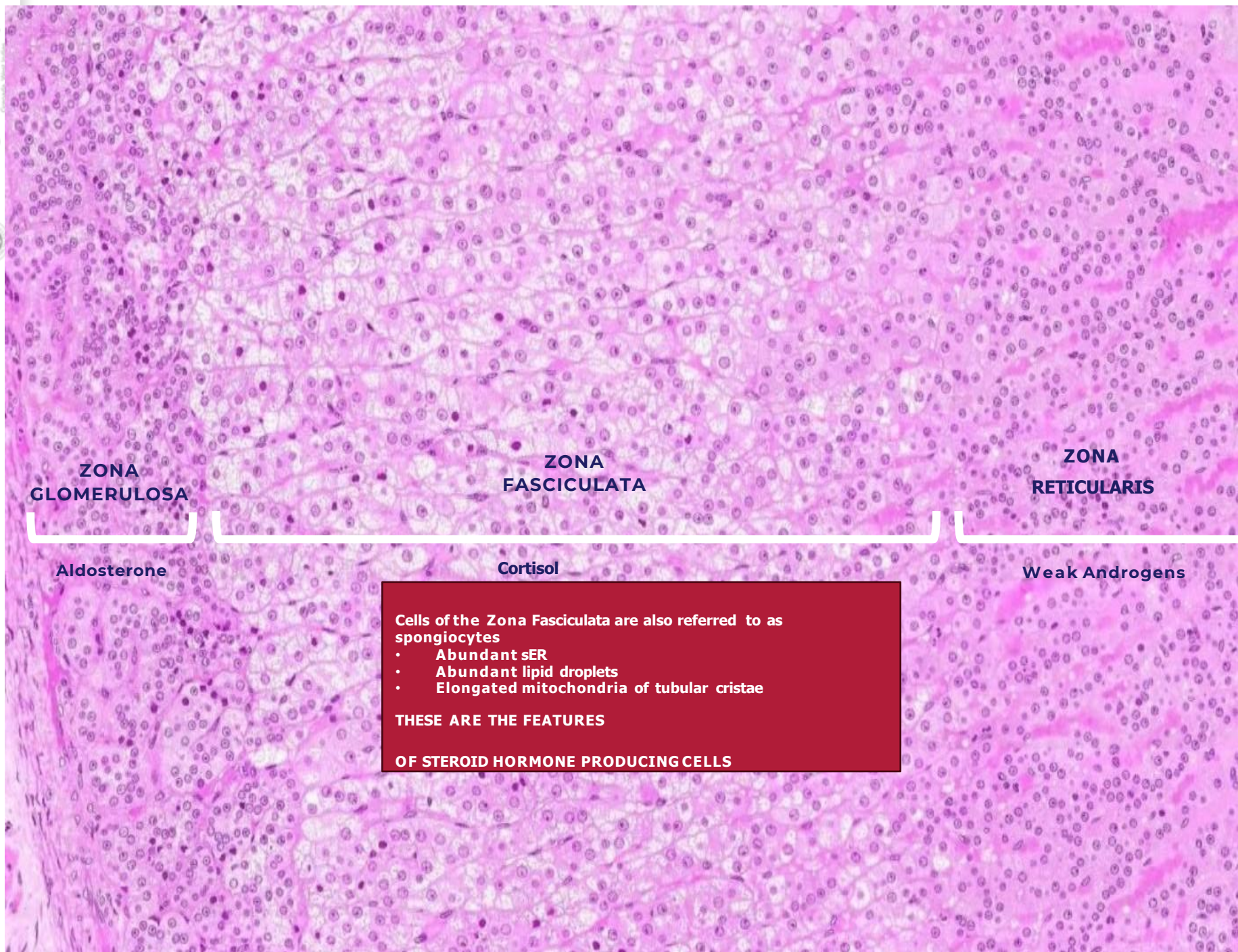


Adrenal Glands

Adrenal Cortex

- **Zona Glomerulosa (15% of Cortex)**
 - Closely packed ovoid clusters arranged in curved columns
 - 1 - 2 layers thick
 - **Aldosterone** (Renin Angiotensin Aldosterone System) → Angiotensin II stimulates Z.G.
- **Zona Fasciculata (80% of Cortex)**
 - 1-2 cells layers thick. Large polyhedral cells called spongiocytes arranged in linear cords
 - ✓ Several lipid droplets = foamy appearance
 - ✓ Pale spherical nuclei
 - ✓ Well-developed smooth ER
 - ✓ Mitochondria with tubular cristae
 - Produce Cortisol mainly under the influence of ACTH
- **Zona Reticularis (5 -7% of Cortex)**
 - Irregular cords of small cells
 - Mainly weak androgens and some cortisol

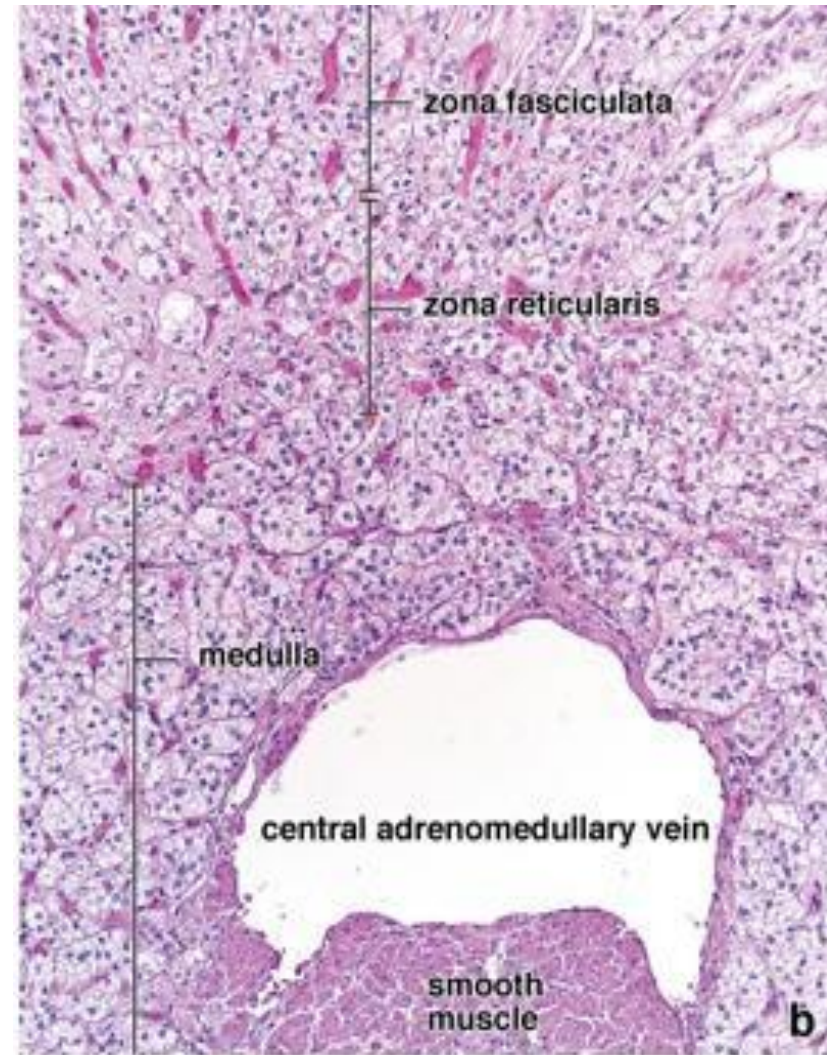




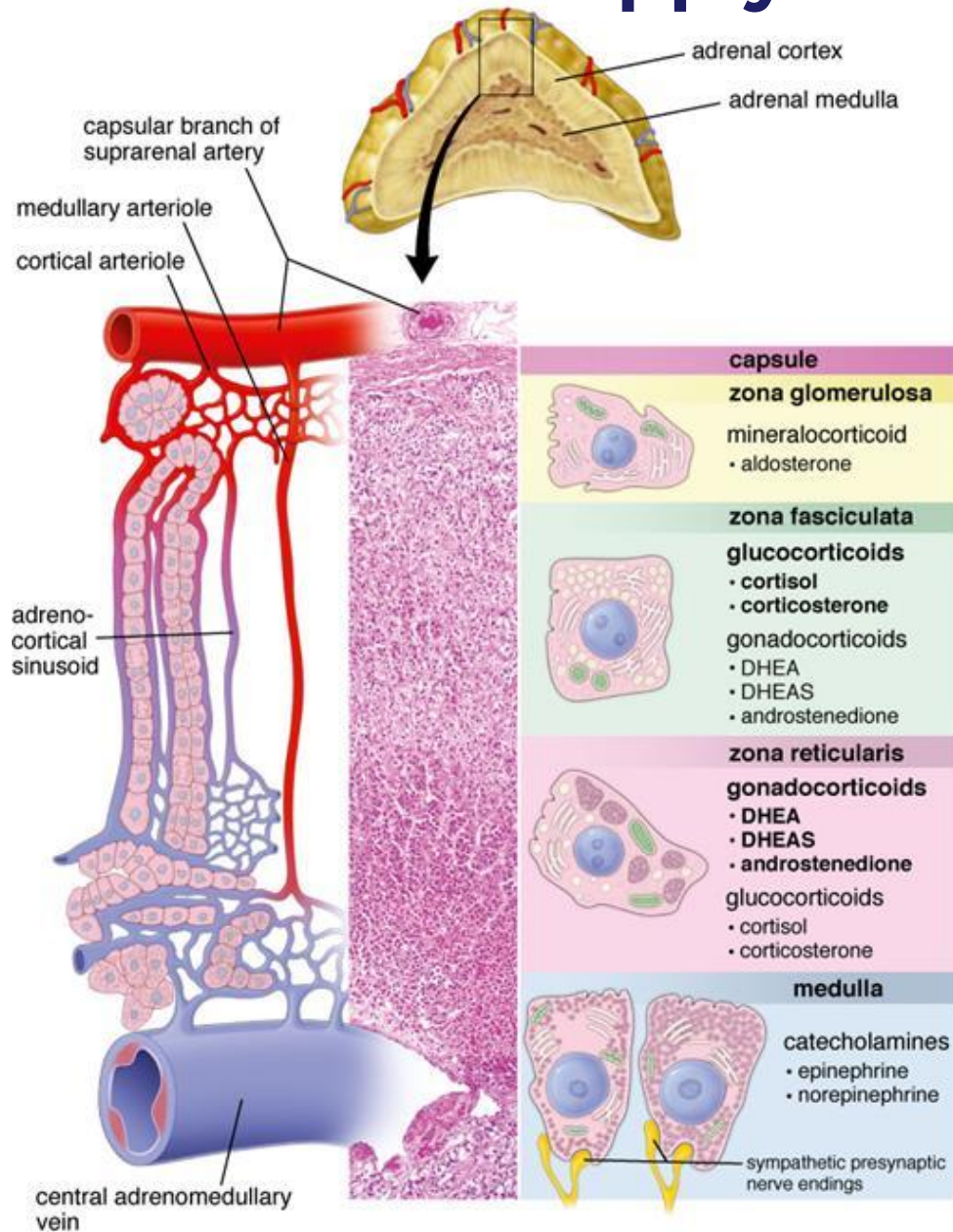
Adrenal Glands

Adrenal Medulla

- **Chromaffin Cells** (Medullary cells) – pale staining epithelioid cells. Modified neurons.
 - Produce catecholamines epinephrine and norepinephrine.
 - Intensifies sympathetic responses.
- **Ganglion cells** - axons extend to cortex to modulate cortex secretion and innervate blood vessels. Can also be a part of splanchnic nerves
- **Large Central Adrenomedullary Vein** (Muscular [medium sized] Vein)



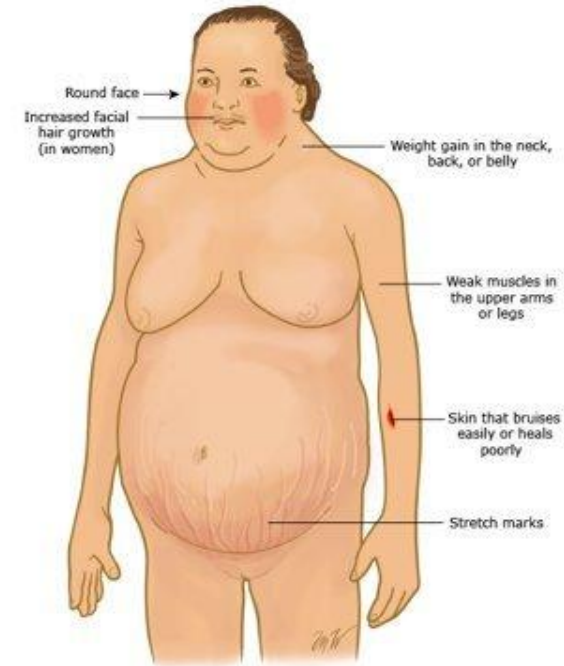
Adrenal Gland Blood Supply



Cushing's Disease VS Cushing's Syndrome

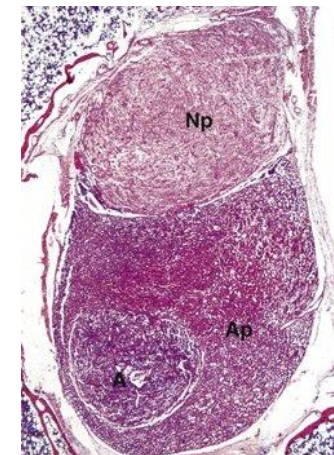
Cushing's Syndrome "Peripheral Dz."

- Too much cortisol/ Glucocorticoid (usually low ACTH Levels).
 - (1) Tumor of the Adrenal Gland,
 - (2) Exogenous use of cortisol
 - (3) *Can also be caused by a non-pituitary ACTH secreting tumor*



Cushing's Disease "Central Disease"

- Pituitary Adenoma producing **too much ACTH** leading to increase glucocorticoid (Cortisol) production and secretion from Zona Fasciculata.
- Microscopic Findings: Diffuse Cortex hyperplasia; uniformly and regularly thickened often by cells of one type (Z.F.)



Low Magnification Light Micrograph Human Pituitary Gland with **Corticotrophic Adenoma** (Basophilic Mass of Cells-A). [AP] Adenohypophysis; [NP] Neurohypophysis

Pancreas

Anatomy

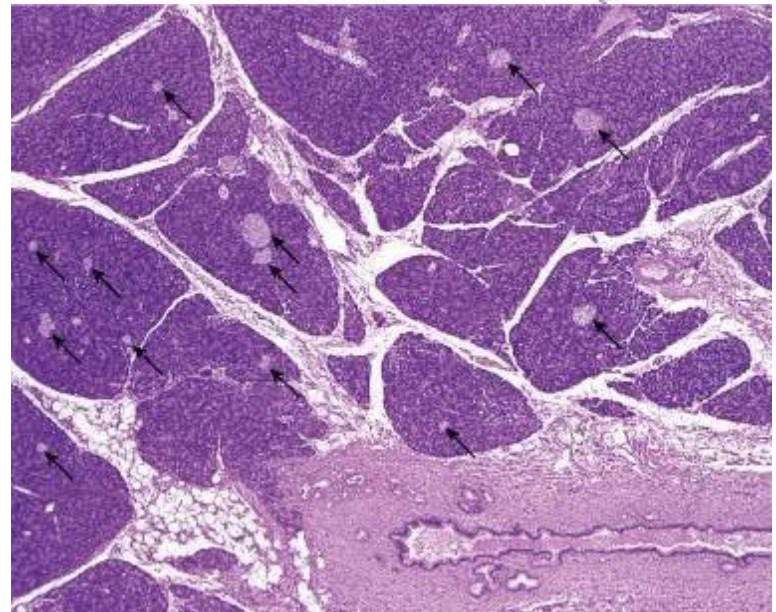
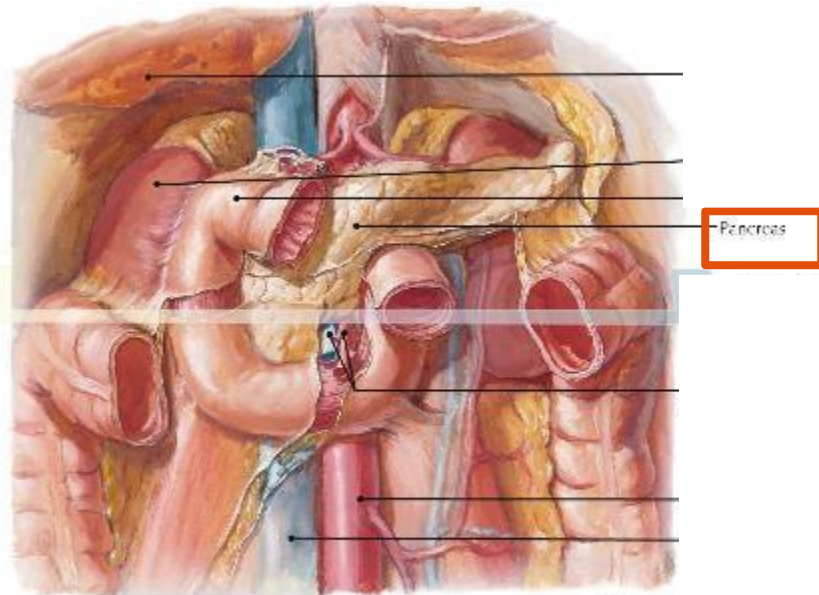
- Retroperitoneal organ with four regions: head
- neck, body, and tail
- Blood supply from the branches of the celiac trunk and superior mesenteric artery.
- Derived from the foregut.

Histology:

- Exocrine Pancreas has acini with granular appearance dark staining with intense basophilia on H&E
- Endocrine Pancreas pale staining on H&E

Function

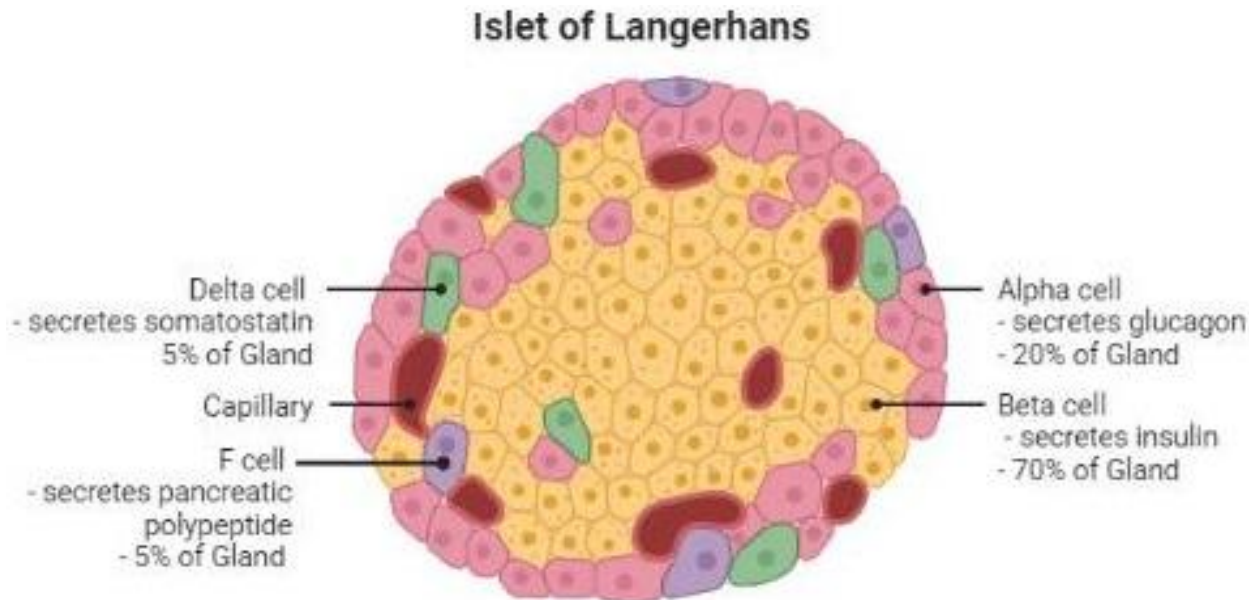
- **Dual endocrine and exocrine function.**
- Endocrine function via the **Islets of Langerhans**.



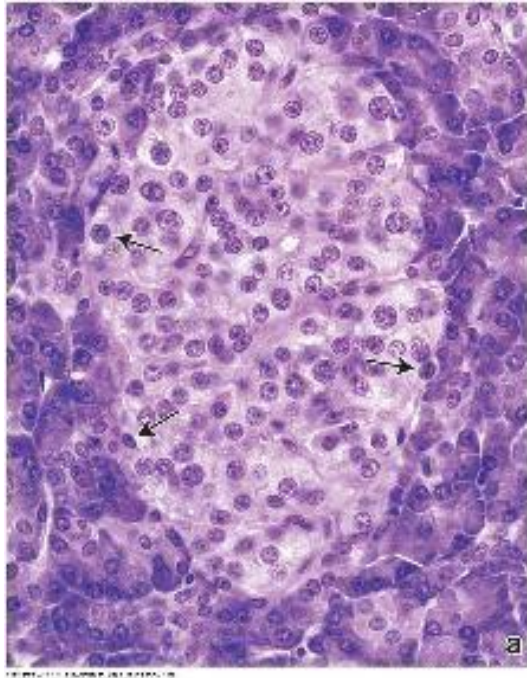
Low Magnification light micrograph H&E Stain Pancreas arrows point to Islet of Langerhans

Pancreas

- **Alpha (A) cells:** Glucagon: (increases blood glucose)
- **Beta (B) Cells:** Insulin (decreases blood glucose)
- **Delta (D) Cells:** Somatostatin inhibits endocrine secretion.
- **F Cells or PP (Pancreatic Polypeptide) Cells:** Stimulates gastric chief cells. Inhibits bile secretion, intestinal motility, pancreatic enzymes and Bicarb secretion)



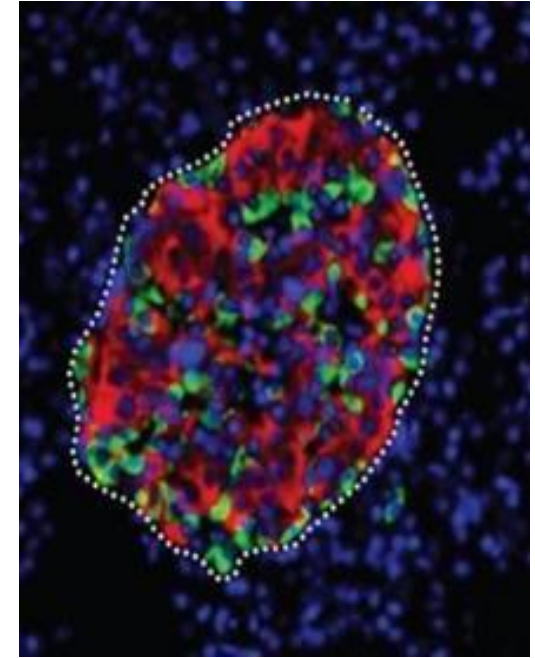
Pancreas



A. High magnification light micrograph
H&E Stain Pancreas Islets of Langerhans.
Arrow Points to A cells.



B. High magnification light micrograph Special
Stain Grimelius silver Island of Langerhans
Rat Pancreas



Immunofluorescent Stain Islets of Langerhans
Green cells → Alpha Cells; Red Cells → Beta
Cells

Resources

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